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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 10

NUMERICAL METHODS

SUGGESTED TIME: 3 PERIODS

IN THIS CHAPTER

This chapter develops the three core numerical tools of the GCE Further Mathematics syllabus: Simpson's rule for definite integrals, Taylor series for polynomial approximation of functions, and step-by-step methods (Euler and improved Euler) for differential equations. Emphasis is placed on accuracy, error awareness, and fluent exam technique.

- By the end of this chapter I can state and apply Simpson's rule with an even number of strips to approximate a definite integral.
- By the end of this chapter I can set up an ordinate table from a formula or from given data and compute the integral estimate accurately.
- By the end of this chapter I can compare Simpson's estimate with the trapezium rule and with exact integration, and comment on accuracy.
- By the end of this chapter I can derive the Maclaurin and Taylor series of standard functions (e^x, sin x, cos x, ln(1+x), (1+x)^n).
- By the end of this chapter I can use a Taylor polynomial to approximate function values and estimate the error involved.
- By the end of this chapter I can use series to evaluate limits and approximate integrals that have no elementary antiderivative.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Simpson's Rule
2. Taylor and Maclaurin Series
3. Applications of Taylor Approximations
4. Step-by-Step Methods for Differential Equations
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

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- By the end of this chapter I can use a Taylor polynomial to approximate function values and estimate the error involved.
- By the end of this chapter I can use series to evaluate limits and approximate integrals that have no elementary antiderivative.
- By the end of this chapter I can apply Euler's step-by-step method to approximate the solution of a first-order differential equation with a given initial condition.
- By the end of this chapter I can apply the improved Euler (Heun) method and compare its accuracy with Euler's method.
- By the end of this chapter I can interpret numerical solutions graphically and relate them to the exact solution when it exists.

PREREQUISITES

- Definite integration and the trapezium rule (Lower Sixth)
- Differentiation of standard functions and higher-order derivatives
- First-order differential equations with separable variables (Lower Sixth)
- Sigma notation, sequences and basic limits
- Use of calculators and tabular organisation of data

1 SIMPSON'S RULE

An engineer at the Douala seaport must estimate the cross-sectional area of an irregularly dredged channel from depth soundings taken every 5 metres — no simple formula exists, so she turns to Simpson's rule. Meanwhile, a technician approximates sin(0.2) for a load calculation without a calculator, using a Taylor polynomial. And a health officer modelling the spread of an infection in Yaoundé solves a differential equation step by step when no exact solution is available. All three problems share one idea: when exact answers are out of reach, **numerical methods** give accurate, controlled approximations. In this chapter we master three of the most powerful of these tools. We begin with the first: a beautifully simple rule for estimating areas and integrals, named after Thomas Simpson.

1.1 From the Trapezium Rule to a Parabolic Approximation

You already know the **trapezium rule**: to estimate $\int_a^b f(x)\,dx$, divide $[a, b]$ into n strips of equal width $h = \dfrac{b-a}{n}$, join the tops of consecutive ordinates by straight line segments (chords), and add the areas of the resulting trapezia:

$$\int_a^b f(x)\,dx \approx \frac{h}{2}\left[y_0 + y_n + 2\left(y_1 + y_2 + \cdots + y_{n-1}\right)\right].$$

WARNING **Why the trapezium rule is limited**

A straight chord is a poor model of a curved graph. If the curve bends away from the chord, every strip carries an error of the same sign, and the errors accumulate. For a curve with strong curvature, the trapezium rule needs a very large number of strips to be accurate. The natural remedy: replace the straight chord by a curve that bends — the simplest such curve is a **parabola**.

The key observation is this: a parabola $y = px^2 + qx + r$ has three coefficients, so it can be made to pass exactly through three points. Take the ordinates in threes — $(x_0, y_0), (x_1, y_1), (x_2, y_2)$ — fit a parabola through them, and compute the exact area under that parabola. That area approximates the area under the true curve over the first pair of strips. Repeat over each pair of strips and add.

1.2 Derivation: Area Under One Pair of Strips

EXPERIMENT **Fitting a parabola through three equally spaced points**

Consider three equally spaced ordinates y_0, y_1, y_2 at $x = -h,\ 0,\ h$ (we may centre them at 0 without loss of generality, since area is unchanged by a horizontal shift). Fit $y = px^2 + qx + r$:

$$y_0 = ph^2 - qh + r, \qquad y_1 = r, \qquad y_2 = ph^2 + qh + r.$$

Adding the first and third equations: $y_0 + y_2 = 2ph^2 + 2r$, so $ph^2 = \dfrac{y_0 - 2y_1 + y_2}{2}$ since $r = y_1$. Now integrate the parabola from $-h$ to h:

$$A_{\text{pair}} = \int_{-h}^{h} (px^2 + qx + r)\,dx = \left[\frac{px^3}{3} + \frac{qx^2}{2} + rx\right]_{-h}^{h} = \frac{2ph^3}{3} + 2rh$$
$$= \frac{2h}{3}\cdot\frac{y_0 - 2y_1 + y_2}{2} + 2hy_1 = \frac{h}{3}(y_0 - 2y_1 + y_2 + 6y_1)$$
$$= \frac{h}{3}(y_0 + 4y_1 + y_2).$$

Notice that the coefficient q has disappeared: the area depends only on the three ordinates. This is the famous **one–four–one** pattern for a pair of strips.

1.3 Statement of Simpson's Rule

Now divide $[a, b]$ into an **even** number n of strips of width $h = \dfrac{b-a}{n}$, with ordinates $y_0, y_1, \dots, y_n$. Apply the pair result to strips $(0, 1, 2)$, then $(2, 3, 4)$, and so on:

$$A \approx \frac{h}{3}(y_0 + 4y_1 + y_2) + \frac{h}{3}(y_2 + 4y_3 + y_4) + \cdots + \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n).$$

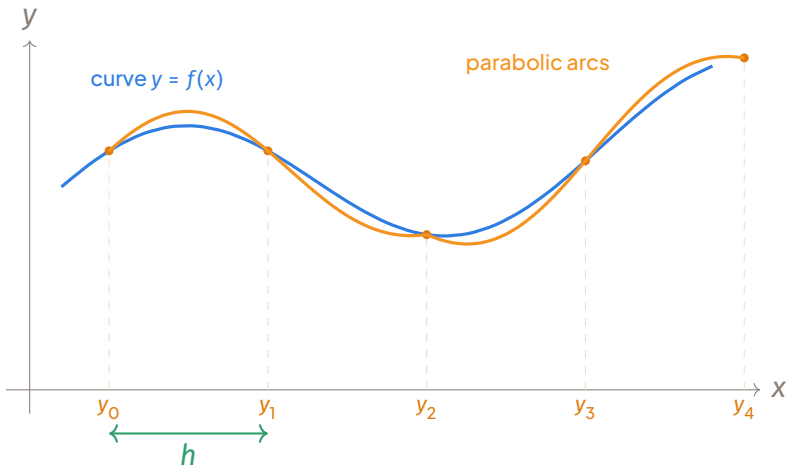
The interior ordinates $y_2, y_4, \dots$ appear in two pairs, hence with coefficient 2; the ordinates $y_1, y_3, \dots$ appear once with coefficient 4; the end ordinates y_0, y_n appear once with coefficient 1.

DEFINITION **Simpson's Rule**

Let f be continuous on $[a, b]$, divided into an **even** number n of strips of equal width $h = \dfrac{b-a}{n}$, with ordinates $y_k = f(a + kh)$, $k = 0, 1, \dots, n$. Then

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\Big[y_0 + y_n + 4\sum_{\text{odd } k} y_k + 2\sum_{\substack{\text{even } k \\ k \neq 0, n}} y_k\Big]$$

The coefficients follow the pattern 1, 4, 2, 4, 2, …, 2, 4, 1.



Four strips ($n = 4$): a parabola is fitted through (y_0, y_1, y_2) and another through (y_2, y_3, y_4).

METHOD **Applying Simpson's Rule: the 1–4–2–4–… –1 pattern**

1. Check the conditions: n must be even and all strips of equal width $h = \dfrac{b-a}{n}$.
2. Build the ordinate table: compute $y_k = f(a+kh)$ for $k = 0, 1, \ldots, n$ (or read the values from experimental data), keeping at least 4 decimal places.
3. Classify the ordinates starting the count from y_0: odd subscripts ($y_1, y_3, \ldots$) get coefficient 4; even subscripts in between ($y_2, y_4, \ldots$) get coefficient 2; the two ends get coefficient 1.
4. Apply the formula: $A \approx \dfrac{h}{3}\left[y_0 + y_n + 4(\text{odd sum}) + 2(\text{even sum})\right]$.
5. Round sensibly and state the units.

WARNING **Count from y_0, not from 1!**

The most frequent error at the GCE is to call y_1 the "first" ordinate and give it coefficient 1. The subscript decides: $y_1, y_3, y_5, \ldots$ (odd subscripts) take 4; $y_2, y_4, \ldots$ (even subscripts, excluding the ends) take 2. Write the pattern $1, 4, 2, 4, \ldots, 4, 1$ above your table before computing.

WARNING **Even number of strips, equal widths**

Simpson's rule in this form is only valid when (i) the number of strips n is even (so the ordinates group into complete triples) and (ii) all strips have the same width h. With an odd number of strips, one strip is left over; with unequal spacing, the 1–4–1 derivation collapses. If data come at irregular intervals, you must use the trapezium rule strip by strip instead.

1.4 Worked Applications

WORKED EXAMPLE 1 **Area under a curve**

Use Simpson's rule with 4 strips to estimate $\displaystyle\int_0^2 \frac{1}{1+x^2}\,dx$, giving 4 decimal places. Compare with the exact value.

SOLUTION

Solution. $n = 4$ (even ✓), $h = \dfrac{2-0}{4} = 0.5$. Ordinates:

k	0	1	2	3	4
x_k	0	0.5	1.0	1.5	2.0
y_k	1.0000	0.8000	0.5000	0.3077	0.2000
coeff.	1	4	2	4	1

$$A \approx \frac{0.5}{3}\big[1.0000+0.2000+4(0.8000+0.3077)+2(0.5000)\big] = \frac{0.5}{3}\big[1.2000+4.4308+1.0000\big] = \frac{0.5}{3}(6.6308).$$

Hence $A \approx 1.1051$. The exact value is $\big[\arctan x\big]_0^2 = \arctan 2 \approx 1.1071$. The error is about 0.002 — remarkably small for only 4 strips. (The trapezium rule with the same data gives 1.1038, less accurate.)

WORKED EXAMPLE 2 **Distance from a velocity–time record**

A lorry travelling from Douala to Edéa has its speed recorded every 10 seconds:

t (s)	0	10	20	30	40	50	60
v (m/s)	0	6.2	10.5	13.1	14.0	13.6	12.8

Estimate the distance covered in the first 60 seconds.

SOLUTION

Solution. Distance $= \int_0^{60} v\,dt$. Here $n = 6$ (even ✓), $h = 10$. Coefficients: 1, 4, 2, 4, 2, 4, 1.

$$s \approx \frac{10}{3}\big[0 + 12.8 + 4(6.2 + 13.1 + 13.6) + 2(10.5 + 14.0)\big]$$
$$= \frac{10}{3}\big[12.8 + 4(32.9) + 2(24.5)\big] = \frac{10}{3}\big[12.8 + 131.6 + 49.0\big] = \frac{10}{3}(193.4).$$

Hence $s \approx 644.7$ m, i.e. about 645 m.

WORKED EXAMPLE 3 **The Douala channel cross-section**

Depth soundings across a dredged channel of width 40 m, taken every 5 m, give depths (in metres):

x (m)	0	5	10	15	20	25	30	35	40
depth d (m)	0.0	3.1	5.2	6.4	6.8	6.1	4.7	2.6	0.0

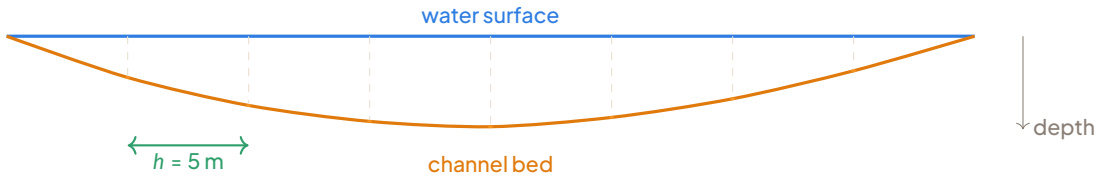
Estimate the cross-sectional area.

SOLUTION

Solution. $n = 8$ (even ✓), $h = 5$. Odd-subscript depths: $3.1 + 6.4 + 6.1 + 2.6 = 18.2$; even-subscript (interior): $5.2 + 6.8 + 4.7 = 16.7$.

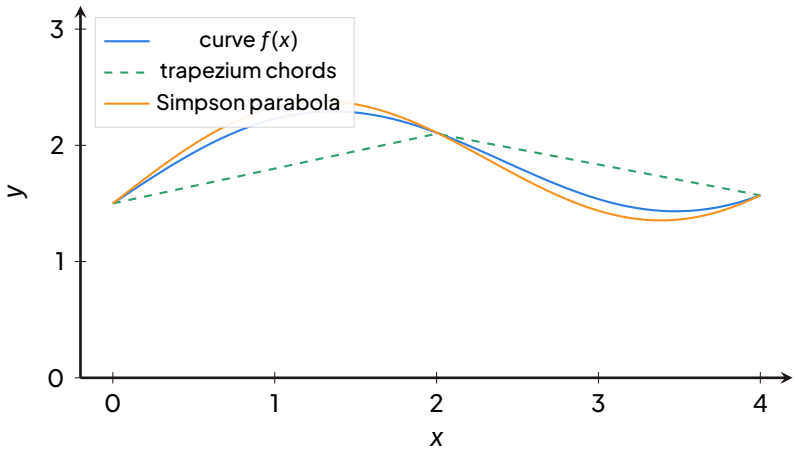
$$A \approx \frac{5}{3}\big[0.0 + 0.0 + 4(18.2) + 2(16.7)\big] = \frac{5}{3}\big[72.8 + 33.4\big] = \frac{5}{3}(106.2) = 177.0\ \text{m}^2.$$

The cross-sectional area is approximately **177 m²** — exactly the figure the port engineer needs to plan the dredging.



Cross-section of the dredged channel at Douala: equally spaced depth soundings every 5 m.

1.5 Trapezium versus Simpson; Exactness for Cubics



The parabola (orange) hugs the curve far better than the chords (green): Simpson's rule is much more accurate for the same ordinates.

PROPERTY **Simpson's rule is exact for polynomials of degree at most 3**

If f is a polynomial of degree ≤ 3, then Simpson's rule gives the exact value of $\int_a^b f(x)\,dx$, for any even n. (Enrichment — the proof is not examinable, but the fact is worth knowing.)

Why? (enrichment). For one pair of strips centred at 0, direct integration shows both $\int_{-h}^{h} x^3\,dx = 0$ and $\frac{h}{3}[(-h)^3 + 4(0)^3 + h^3] = 0$: the rule is exact for x^3 (and trivially for 1, x, x^2 by construction). By linearity, it is exact for every cubic. This explains the striking accuracy seen in our worked example: the error term involves the fourth derivative of f, so for smooth, gently curving functions the error is tiny. In fact one can show the error behaves like Kh^4 (compared with Kh^2 for the trapezium rule): halving h divides Simpson's error by about 16, but the trapezium error only by 4.

1.6 Choosing the Number of Strips

KEY POINTS **Accuracy and the choice of n**

- n must be even; $h = \frac{b-a}{n}$ must be constant.
- Doubling the number of strips divides Simpson's error by roughly **16** — a cheap way to gain accuracy.

- In practice, compute with n and with $2n$ strips: if the two estimates agree to the required number of decimal places, the answer is reliable.
- More strips mean more computation and more rounding; keep at least 4–5 decimal places in the ordinates.

EXERCISE 1 **Estimating an integral**

Use Simpson's rule with 6 strips to estimate $\int_0^3 e^{-x^2}\,dx$, giving your answer to 4 decimal places.

SOLUTION **Exercise 1**

$h = \frac{3-0}{6} = 0.5$. Ordinates $y_k = e^{-x_k^2}$:

x_k	0	0.5	1.0	1.5	2.0	2.5	3.0
y_k	1.0000	0.7788	0.3679	0.1054	0.0183	0.0019	0.0001
coeff.	1	4	2	4	2	4	1

$$A \approx \frac{0.5}{3}\left[1.0000+0.0001+4(0.7788+0.1054+0.0019)+2(0.3679+0.0183)\right] = \frac{0.5}{3}\left[1.0001+3.5444+0.7724\right].$$

$A \approx \frac{0.5}{3}(5.3169) \approx 0.8862$. (The true value is about 0.8862 — excellent agreement.)

EXERCISE 2 **Soundings on the Wouri**

The width of a river channel is 24 m. Depths (m) measured every 3 m are: 0, 1.8, 3.0, 3.7, 3.5, 2.8, 1.9, 0.9, 0. Use Simpson's rule to estimate the cross-sectional area.

SOLUTION **Exercise 2**

$n = 8$, $h = 3$. Odd sum: $1.8 + 3.7 + 2.8 + 0.9 = 9.2$; even sum: $3.0 + 3.5 + 1.9 = 8.4$.

$$A \approx \frac{3}{3}\left[0 + 0 + 4(9.2) + 2(8.4)\right] = 36.8 + 16.8 = 53.6\ \text{m}^2.$$

DID YOU KNOW? **Thomas Simpson**

Thomas Simpson (1710–1761) was a self-taught English mathematician who worked as a weaver before becoming professor at the Royal Military Academy. The rule bearing his name was in fact known a century earlier to Bonaventura Cavalieri — a reminder that in mathematics, ideas often travel further than their inventors' names. Today, the same 1–4–2–4–… –1 pattern runs silently inside every calculator and every ship-design and harbour-engineering program in the world, including those used at the Port of Douala.

2 TAYLOR AND MACLAURIN SERIES

In the previous section, Simpson's Rule showed us the power of a simple idea: replace a complicated curve by a polynomial (a parabola) that is easy to work with, and the computation becomes almost trivial. We now push exactly the same idea much further. Instead of approximating a curve over an interval by a parabola, we ask: can we approximate a smooth function near a single point by a polynomial of any degree we like? The answer is the **Taylor series**, one of the most powerful tools in all of mathematics — it is how your calculator actually computes $\sin x$, e^x and $\ln x$.

2.1 The idea: matching derivatives at a point

EXPERIMENT **Building a polynomial that imitates a function**

Suppose we want to approximate $f(x) = e^x$ near $x = 0$ by a polynomial

$$P(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + \cdots$$

The best we can ask is that P and f "agree as much as possible" at $x = 0$. What can they agree on?

- Same value: $P(0) = f(0)$ gives $c_0 = f(0) = 1$.
- Same slope: $P'(0) = f'(0)$ gives $c_1 = f'(0) = 1$.
- Same curvature: $P''(0) = f''(0)$ gives $2c_2 = f''(0) = 1$, so $c_2 = \frac{1}{2}$.
- Same third derivative: $P'''(0) = f'''(0)$ gives $6c_3 = 1$, so $c_3 = \frac{1}{6}$.

Each extra condition forces one more coefficient, and the polynomial hugs the curve more and more closely. The pattern $c_k = \dfrac{f^{(k)}(0)}{k!}$ is already visible — this is the heart of the Taylor series.

The intuition is geometric: a constant matches the position of the curve, a line matches position

and direction, a parabola also matches curvature, and so on. The more derivatives we match at $x = a$, the better the polynomial imitates f near a.

2.2 The Taylor series

DEFINITION **Taylor series of f about $x = a$**

Let f be a function whose derivatives of all orders exist at $x = a$. The **Taylor series** of f about $x = a$ is

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \cdots + \frac{h^n}{n!}f^{(n)}(a) + \cdots$$

Equivalently, putting $x = a + h$,

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \cdots$$

The polynomial obtained by stopping after the term in $(x-a)^n$ is called the **Taylor polynomial of degree n**, denoted $T_n(x)$ (or $P_n(x)$).

PROPERTY **Why the factorials appear (derivation)**

Assume $f(x)$ can be written as a power series centred at a:

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n + \cdots$$

Setting $x = a$ kills every term except the first: $c_0 = f(a)$. Differentiate term by term:

$$f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots$$

and setting $x = a$ gives $c_1 = f'(a)$. Differentiating again,

$$f''(x) = 2c_2 + 3\cdot 2\,c_3(x-a) + \cdots \quad\Longrightarrow\quad c_2 = \frac{f''(a)}{2}.$$

After k differentiations, the constant term is $k!\,c_k = f^{(k)}(a)$, because differentiating $(x-a)^k$ exactly k times produces $k!$. Hence

$$c_k = \frac{f^{(k)}(a)}{k!},$$

which is precisely the Taylor coefficient. The factorial $k!$ is therefore not a decoration — it comes from repeated differentiation of powers.

METHOD **Computing a Taylor polynomial by successive differentiation**

To find the Taylor polynomial of degree n for f about $x = a$:

1. Compute the derivatives $f'(x), f''(x), \ldots, f^{(n)}(x)$.
2. Evaluate each derivative at $x = a$: $f(a), f'(a), \ldots, f^{(n)}(a)$.
3. Divide the k-th value by $k!$ to get the coefficient of $(x - a)^k$.
4. Assemble: $T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$.

2.3 Taylor's theorem with remainder (Lagrange form)

PROPERTY **Taylor's theorem with remainder**

If f has derivatives up to order $n + 1$ on an interval containing a and x, then

$$f(x) = T_n(x) + R_n(x)$$

where $T_n(x)$ is the Taylor polynomial of degree n and the **Lagrange remainder** is

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - a)^{n+1}$$

for some ξ between a and x. This gives an exact expression for the error when approximating $f(x)$ by $T_n(x)$.

KEY POINTS **Using the remainder for error bounds**

If we can bound $|f^{(n+1)}(t)| \le M$ for all t between a and x, then

$$|R_n(x)| \le \frac{M}{(n+1)!}|x - a|^{n+1}.$$

This is the standard way to guarantee the accuracy of a Taylor approximation in examinations.

2.4 The Maclaurin series and standard expansions

DEFINITION **Maclaurin series**

The **Maclaurin series** is the Taylor series about $a = 0$:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$

EXEMPLE **Deriving the Maclaurin series of e^x**

Let $f(x) = e^x$. Every derivative is again e^x, so $f^{(k)}(0) = e^0 = 1$ for all k. Therefore

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

valid for every real x. Setting $x = 1$ gives the famous identity $e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots$

EXEMPLE **Deriving the Maclaurin series of $\sin x$**

Let $f(x) = \sin x$. The derivatives cycle with period 4:

$$\sin x \to \cos x \to -\sin x \to -\cos x \to \sin x \to \cdots$$

At $x = 0$ the values cycle through $0, 1, 0, -1, 0, 1, \ldots$ So only odd powers survive, with alternating signs:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Notice the series contains only odd powers — exactly as expected, since $\sin x$ is an odd function.

EXEMPLE **Deriving the Maclaurin series of $\cos x$**

Let $f(x) = \cos x$. Derivatives cycle: $\cos x \to -\sin x \to -\cos x \to \sin x \to \cos x$. At $x = 0$: $1, 0, -1, 0, 1, \ldots$ Only even powers survive:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

KEY POINTS **Standard Maclaurin expansions and their intervals of validity**

Function	Maclaurin series	Valid for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	all $x \in \mathbb{R}$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	all $x \in \mathbb{R}$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	all $x \in \mathbb{R}$
$\ln(1 + x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$-1 < x \leq 1$
$(1 + x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$	$\lvert x \rvert < 1$

The last expansion is the **binomial series**; when n is a positive integer it terminates and reduces to the ordinary binomial theorem.

WARNING **Forgetting factorials or powers of $(x - a)$**

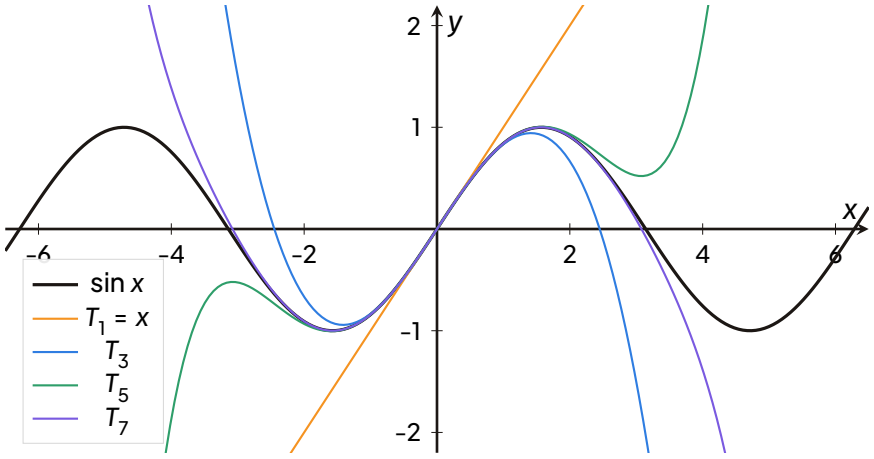
The two most frequent errors in the examination are:

- Writing the coefficient of x^3 in $\sin x$ as $-\frac{1}{3}$ instead of $-\frac{1}{3!} = -\frac{1}{6}$. The denominator is $k!$, not k.
- Expanding about $x = a$ but writing powers of x instead of powers of $(x-a)$. For example, the Taylor series of e^x about $x = 1$ is $e\left[1 + (x - 1) + \frac{(x-1)^2}{2!} + \cdots\right]$, with $(x - 1)$ everywhere.

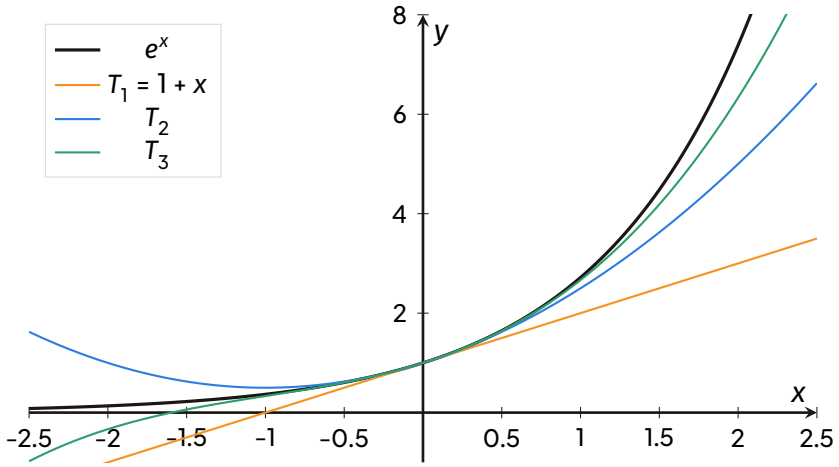
A quick check: setting $x = a$ in your polynomial must give $f(a)$ exactly.

2.5 Seeing the convergence

The figures below show how the Maclaurin polynomials wrap themselves around the true curve as the degree increases.



Maclaurin polynomials of degrees 1, 3, 5, 7 approximating $\sin x$. Each extra pair of terms extends the range where the approximation is accurate.



Truncations of orders 1, 2 and 3 of the Maclaurin series of e^x near the origin. Close to $x = 0$ even T_2 is excellent; far away, more terms are needed.

2.6 Building new series by substitution and multiplication

We rarely need to differentiate from scratch. New series are obtained from the standard ones.

EXEMPLE **Substitution: e^{-x^2} and $\sin 2x$**

Replace x by $-x^2$ in the exponential series:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

Replace x by $2x$ in the sine series:

$$\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!} - \cdots = 2x - \frac{4x^3}{3} + \frac{4x^5}{15} - \cdots$$

EXEMPLE **Multiplication by a power:** $x\cos x$

Multiply the cosine series through by x:

$$x\cos x = x\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) = x - \frac{x^3}{2!} + \frac{x^5}{4!} - \cdots$$

EXEMPLE **Division:** $\dfrac{\sin x}{x}$

For $x \neq 0$, divide the sine series by x:

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots$$

This shows $\lim\limits_{x\to 0} \dfrac{\sin x}{x} = 1$ immediately.

EXEMPLE **Composition:** $e^{\sin x}$ **up to** x^3

Substitute the series of $\sin x$ into the exponential series, keeping terms up to x^3:

$$e^{\sin x} = 1 + \sin x + \frac{\sin^2 x}{2!} + \frac{\sin^3 x}{3!} + \cdots$$

$$\sin x = x - \frac{x^3}{6} + O(x^5), \quad \sin^2 x = x^2 + O(x^4), \quad \sin^3 x = x^3 + O(x^5)$$

$$e^{\sin x} = 1 + \left(x - \frac{x^3}{6}\right) + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4) = 1 + x + \frac{x^2}{2} + O(x^4)$$

2.7 Approximating values and estimating the error

WORKED EXAMPLE 4 **Approximating** $\sin(0.2)$

Using the Maclaurin series of $\sin x$, approximate $\sin(0.2)$ and estimate the error.

SOLUTION

Solution. With $x = 0.2$:

$$\sin(0.2) = 0.2 - \frac{0.2^3}{3!} + \frac{0.2^5}{5!} - \cdots$$

First term: 0.2. Second term: $\frac{0.008}{6} \approx 0.001333$. Third term: $\frac{0.00032}{120} \approx 0.0000027$. Taking the first two terms: $\sin(0.2) \approx 0.2 - 0.001333 = 0.198667$.
The series is alternating with decreasing terms, so the error is less than the first neglected term: the error is below 0.0000027, i.e. our value is correct to about 5 decimal places. (The true value is 0.198669 …)

WORKED EXAMPLE 5 **Approximating $e^{0.1}$ and $\ln(1.1)$**

(a) Approximate $e^{0.1}$ using four terms. (b) Approximate $\ln(1.1)$ using three terms and comment on the error.

SOLUTION

Solution. (a) $e^{0.1} \approx 1 + 0.1 + \frac{0.01}{2} + \frac{0.001}{6} = 1 + 0.1 + 0.005 + 0.0001667 = 1.1051667$.
The next term is $\frac{0.1^4}{4!} = \frac{0.0001}{24} \approx 0.0000042$, so the error is of order 4×10^{-6}. (True value: 1.1051709 …)
(b) $\ln(1.1) = \ln(1 + 0.1) \approx 0.1 - \frac{0.01}{2} + \frac{0.001}{3} = 0.1 - 0.005 + 0.000333 = 0.095333$.
The first neglected term is $\frac{0.1^4}{4} = 0.000025$, bounding the error. (True value: 0.0953102 …) Note that $x = 0.1$ lies inside the interval of validity $-1 < x \le 1$, so the use of the series is legitimate.

WORKED EXAMPLE 6 **Error bound using Lagrange remainder for $e^{0.5}$**

Use the Taylor polynomial of degree 3 for e^x about $x = 0$ to approximate $e^{0.5}$ and find a rigorous error bound.

SOLUTION

Solution. $T_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$. For $x = 0.5$:

$$T_3(0.5) = 1 + 0.5 + \frac{0.25}{2} + \frac{0.125}{6} = 1 + 0.5 + 0.125 + 0.020833 = 1.645833.$$

The Lagrange remainder is $R_3(0.5) = \frac{e^{\xi}}{4!}(0.5)^4$ for some $\xi \in (0, 0.5)$. Since $e^{\xi} < e^{0.5} < e^1 < 3$,

$$|R_3(0.5)| < \frac{3}{24} \times 0.0625 = 0.0078125.$$

A sharper bound uses $e^{\xi} < e^{0.5} \approx 1.65$, giving $|R_3| < 0.0043$. The true value is $e^{0.5} \approx 1.64872$, so the actual error is about 0.0029.

WARNING **Using a series outside its interval of validity**

The series for e^x, $\sin x$ and $\cos x$ converge for all x, but $\ln(1+x)$ requires $-1 < x \le 1$ and $(1+x)^n$ requires $|x| < 1$. Substituting $x = 2$ into the logarithm series gives the nonsense

$$\ln 3 \text{ “=” } 2 - 2 + \tfrac{8}{3} - 4 + \cdots$$

whose terms grow without bound. Always check validity before substituting. To compute $\ln 3$ legitimately, one may instead write $\ln 3 = -\ln(1/3) = -\ln\left(1 - \tfrac{2}{3}\right)$, which is valid.

2.8 Applications: limits and integrals

EXEMPLE **Evaluating a limit using series**

Find $\displaystyle\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$.

Substitute the Maclaurin series $e^x = 1 + x + \tfrac{x^2}{2} + \tfrac{x^3}{6} + \cdots$:

$$\frac{e^x - 1 - x}{x^2} = \frac{\left(1 + x + \tfrac{x^2}{2} + \tfrac{x^3}{6} + \cdots\right) - 1 - x}{x^2} = \frac{\tfrac{x^2}{2} + \tfrac{x^3}{6} + \cdots}{x^2} = \frac{1}{2} + \frac{x}{6} + \cdots$$

As $x \to 0$, the limit is $\tfrac{1}{2}$.

EXEMPLE **Approximating an integral: $\int_0^{0.2} e^{-x^2}\,dx$**

The integral $\int e^{-x^2}\,dx$ has no elementary antiderivative. Use the series $e^{-x^2} = 1 - x^2 + \tfrac{x^4}{2} - \tfrac{x^6}{6} + \cdots$ and integrate term by term:

$$\int_0^{0.2} e^{-x^2}\,dx = \left[x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots\right]_0^{0.2}$$

$$= 0.2 - \frac{0.008}{3} + \frac{0.00032}{10} - \cdots \approx 0.2 - 0.002667 + 0.000032 = 0.197365.$$

The first neglected term is $\tfrac{0.2^7}{42} \approx 3 \times 10^{-7}$, so the error is negligible.

2.9 Enrichment: series solutions of differential equations

DID YOU KNOW? **Power series and differential equations**

A powerful technique (a taste of what comes in the next section) is to assume a differential equation has a power series solution $y = \sum_{k=0}^{\infty} c_k x^k$, substitute it into the equation, and match coefficients of each power of x. For example, $y' = y$ with $y(0) = 1$ forces $c_1 = c_0$, $2c_2 = c_1$, $3c_3 = c_2$, …, giving $c_k = \dfrac{1}{k!}$ — the series of e^x appears all by itself! This confirms that e^x is the natural solution of $y' = y$, and shows how series and differential equations are two faces of the same coin.

EXERCISE 3 **Practice**

1. Find the Maclaurin series of $\cos x$ up to the term in x^6 by successive differentiation, and verify it by differentiating the series for $\sin x$ term by term.
2. Use substitution to obtain the first four terms of the Maclaurin series of $\dfrac{1}{1+x^2}$ (hint: binomial series with $n = -1$), and state its interval of validity.
3. Use three terms of an appropriate series to approximate $\sqrt{1.02}$, and estimate the error using the first neglected term.
4. Find the Taylor series of $\ln x$ about $x = 1$ up to the term in $(x-1)^4$.
5. Use the Maclaurin series of $\sin x$ to evaluate $\displaystyle\lim_{x \to 0} \frac{x - \sin x}{x^3}$.
6. Approximate $\int_0^{0.1} \frac{\sin x}{x}\,dx$ using the first three non-zero terms of the series, and estimate the error.

SOLUTION **Exercise 1**

$f(x) = \cos x$: $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$, $f^{(4)}(0) = 1$, $f^{(5)}(0) = 0$, $f^{(6)}(0) = -1$. Hence

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

Differentiating $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$ term by term gives exactly the same series, as expected since $(\sin x)' = \cos x$.

SOLUTION **Exercise 2**

With $n = -1$ and x replaced by x^2 in $(1+u)^{-1} = 1 - u + u^2 - u^3 + \cdots$:

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots, \qquad \text{valid for } |x| < 1.$$

SOLUTION **Exercise 3**

$\sqrt{1.02} = (1+0.02)^{1/2} \approx 1 + \frac{1}{2}(0.02) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(0.02)^2 = 1 + 0.01 - 0.00005 = 1.00995.$

The first neglected term is $\frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(0.02)^3 = \frac{1}{16}(0.02)^3 = 5\times10^{-7}$, so the error is of order 5×10^{-7}: the approximation is accurate to about 6 decimal places. (True value: 1.0099504…)

SOLUTION **Exercise 4**

Let $f(x) = \ln x$, $a = 1$. Then $f(1) = 0$, $f'(x) = x^{-1}$, $f''(x) = -x^{-2}$, $f'''(x) = 2x^{-3}$, $f^{(4)}(x) = -6x^{-4}$, giving at $x = 1$: $1, -1, 2, -6$. Hence

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots$$

which is exactly the series for $\ln(1+u)$ with $u = x - 1$ — a useful consistency check.

SOLUTION **Exercise 5**

$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$, so $x - \sin x = \frac{x^3}{6} - \frac{x^5}{120} + \cdots$.

$$\frac{x - \sin x}{x^3} = \frac{1}{6} - \frac{x^2}{120} + \cdots \xrightarrow[x\to 0]{} \frac{1}{6}.$$

SOLUTION **Exercise 6**

$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots$. Integrate from 0 to 0.1:

$$\int_0^{0.1} \frac{\sin x}{x}\,dx = \left[x - \frac{x^3}{18} + \frac{x^5}{600} - \cdots\right]_0^{0.1} = 0.1 - \frac{0.001}{18} + \frac{0.00001}{600} \approx 0.1 - 0.0000556 + 1.67\times10^{-8} = 0.0999444.$$

The first neglected term is $\frac{0.1^7}{7\times5040} \approx 2.8 \times 10^{-11}$, so the error is far below 10^{-10}.

KEY POINTS **Key points**

- The Taylor series about $x = a$ matches all derivatives of f at a: the coefficient of $(x - a)^k$ is $\frac{f^{(k)}(a)}{k!}$.
- Maclaurin series is the case $a = 0$; learn the five standard expansions with their intervals of validity.
- Taylor's theorem with Lagrange remainder gives a rigorous error bound: $|R_n(x)| \leq \frac{M}{(n+1)!}|x - a|^{n+1}$.
- Build new series by substitution, multiplication, division or composition — do not re-differentiate unless asked.
- For an alternating series with decreasing terms, the error is bounded by the first neglected term.
- Never use $\ln(1 + x)$ or $(1 + x)^n$ outside their intervals of validity.
- Series are powerful tools for evaluating limits and approximating integrals that have no elementary antiderivatives.

3 APPLICATIONS OF TAYLOR APPROXIMATIONS

In the previous section we built the Taylor and Maclaurin machinery: we learned how to expand functions such as e^x, $\sin x$, $\cos x$, $\ln(1 + x)$ and $(1 + x)^n$ into infinite power series. A natural question now arises: what are these series actually good for? The answer is that Taylor series are one of the most powerful **approximation** tools in all of mathematics. Whenever a calculator, a physicist or an engineer needs a numerical value of a function, it is almost always a truncated Taylor series that does the work behind the scenes. In this section we learn to use these series with judgement: to approximate function values to a required accuracy, to justify the famous small-angle approximations, to evaluate tricky limits, to estimate integrals that have no elementary antiderivative, and to compare our results with Simpson's rule.

3.1 Approximating Function Values to a Required Number of Decimal Places

The idea. Suppose we want $\sin(0.2)$ to 4 decimal places. The Maclaurin series gives

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Substituting $x = 0.2$:

$$\sin(0.2) = 0.2 - \frac{0.008}{6} + \frac{0.00032}{120} - \cdots = 0.2 - 0.0013333\ldots + 0.00000266\ldots - \cdots$$

The terms decrease very fast, so a few terms suffice. The key skill is knowing when to stop: we stop when the first neglected term cannot affect the required decimal place.

METHOD **Approximating a value to n decimal places**

1. Expand. Write down the Maclaurin (or Taylor) series of the function, in its interval of validity.
2. Substitute. Replace x by the given value (converting to radians if it is an angle) and compute the terms one by one, keeping at least $n + 2$ decimal places in your working.
3. Check the first neglected term. For an alternating series whose terms decrease in absolute value to 0, the error made by truncating is smaller in absolute value than the first neglected term. Stop as soon as that term is less than $\frac{1}{2} \times 10^{-n}$ (half a unit of the last required decimal place).
4. Round and conclude. Round your accumulated sum to n decimal places and state the answer clearly.

WORKED EXAMPLE 7 **Approximating $e^{0.1}$ to 5 decimal places**

Use the Maclaurin series of e^x to find $e^{0.1}$ correct to 5 decimal places, justifying the number of terms used.

SOLUTION

Solution. The Maclaurin series $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$ is valid for all $x \in \mathbb{R}$. With $x = 0.1$:

$$\begin{aligned} 1 &= 1.0000000 \\ \frac{x}{1!} &= 0.1000000 \\ \frac{x^2}{2!} &= 0.0050000 \\ \frac{x^3}{3!} &= 0.00016667 \\ \frac{x^4}{4!} &= 0.00000417 \\ \frac{x^5}{5!} &= 0.000000083 \end{aligned}$$

The term $\frac{x^5}{5!} \approx 8.33 \times 10^{-8} < \frac{1}{2} \times 10^{-5}$, and all subsequent terms are smaller still, so they

cannot affect the fifth decimal place. Summing the terms up to x^4:

$$e^{0.1} \approx 1 + 0.1 + 0.005 + 0.00016667 + 0.00000417 = 1.10517084$$

Hence $\boxed{e^{0.1} \approx 1.10517}$ to 5 decimal places (the true value is 1.1051709 …).

WARNING **Truncating before the required accuracy is guaranteed**

A very common GCE mistake is to stop after two or three terms "because the terms look small". Small is not enough: you must prove the first neglected term is below half a unit of the last required decimal place. For example, approximating $\sin(0.5)$ by $0.5 - \frac{0.125}{6} = 0.47917$ gives an error of about 2.6×10^{-4} from the neglected $\frac{x^5}{120}$ term — fine for 3 decimal places, but not for 4. Always state the first neglected term and its size explicitly: examiners award a mark for this justification.

3.2 Small-Angle Approximations

Observation. Draw the graphs of $y = \sin x$ and $y = x$ near the origin: they are almost indistinguishable. This is not a coincidence — it is the Maclaurin series at work.

From the series

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots, \qquad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots,$$

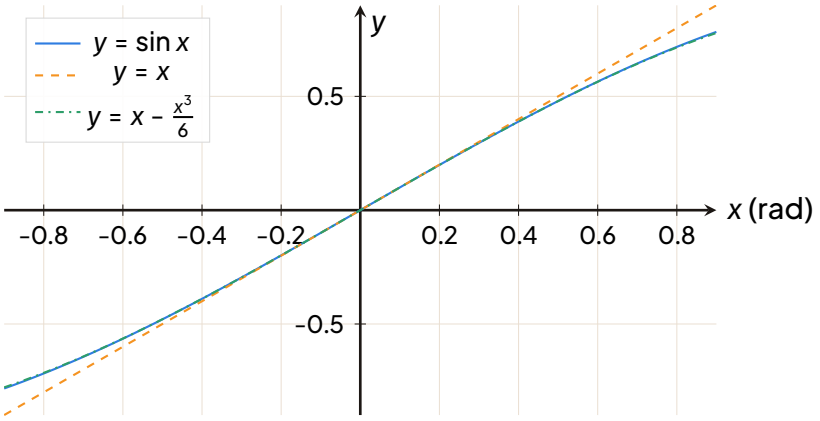
we see immediately that for small x (in radians), the higher powers are negligible:

PROPERTY **Small-angle approximations**

For x small and measured in **radians**:

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad \tan x \approx x.$$

More precisely, the error in $\sin x \approx x$ is about $\frac{x^3}{6}$, the error in $\cos x \approx 1 - \frac{x^2}{2}$ is about $\frac{x^4}{24}$, and the error in $\tan x \approx x$ is about $\frac{x^3}{3}$.



Near $x = 0$, the line $y = x$ is an excellent approximation to $y = \sin x$; the cubic $y = x - \frac{x^3}{6}$ remains accurate much further from the origin.

EXEMPLE **The period of a pendulum**

In physics, the motion of a simple pendulum of length L obeys $\ddot{\theta} = -\frac{g}{L}\sin\theta$. For small oscillations we replace $\sin\theta$ by θ, giving $\ddot{\theta} = -\frac{g}{L}\theta$ — simple harmonic motion with period $T = 2\pi\sqrt{L/g}$. For a pendulum in a Yaoundé laboratory with $L = 1.00\,\text{m}$ and $g = 9.81\,\text{m s}^{-2}$, $T \approx 2\pi\sqrt{1/9.81} \approx 2.01\,\text{s}$. The approximation is excellent provided the amplitude stays below about 10° (0.17 rad), where $\sin\theta$ and θ differ by less than 0.5%.

WARNING **Mixing degrees and radians**

The approximations $\sin x \approx x$ and $\cos x \approx 1 - \frac{x^2}{2}$ are valid only when x is in radians. For example, $\sin 5^\circ$ is not approximately 5! You must first convert: $5^\circ = \frac{5\pi}{180} \approx 0.0873$ rad, so $\sin 5^\circ \approx 0.0873$ (true value 0.0872). Before applying any small-angle approximation, check the unit and convert with $\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$.

3.3 Evaluating Limits Using Series

Many limits of the form $\frac{0}{0}$ become transparent once the numerator is expanded as a series. The technique is: expand, simplify, then let $x \to 0$.

WORKED EXAMPLE 8 **Two classical limits**

Using Maclaurin series, evaluate: (a) $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$ (b) $\lim_{x\to 0}\dfrac{e^x-1-x}{x^2}$.

SOLUTION

Solution. (a) Since $\cos x = 1-\dfrac{x^2}{2}+\dfrac{x^4}{24}-\cdots$,

$$\frac{1-\cos x}{x^2}=\frac{\frac{x^2}{2}-\frac{x^4}{24}+\cdots}{x^2}=\frac{1}{2}-\frac{x^2}{24}+\cdots\xrightarrow[x\to 0]{}\frac{1}{2}.$$

(b) Since $e^x = 1+x+\dfrac{x^2}{2}+\dfrac{x^3}{6}+\cdots$,

$$\frac{e^x-1-x}{x^2}=\frac{\frac{x^2}{2}+\frac{x^3}{6}+\cdots}{x^2}=\frac{1}{2}+\frac{x}{6}+\cdots\xrightarrow[x\to 0]{}\frac{1}{2}.$$

Note how the series method replaces a delicate indeterminate form by simple algebra: the terms that caused the $0/0$ cancel exactly.

3.4 Approximating Integrals with No Elementary Antiderivative

Some innocent-looking integrals, such as $\int e^{-x^2}\,dx$ and $\int \dfrac{\sin x}{x}\,dx$, have no antiderivative expressible with elementary functions. Yet these integrals are essential (the first underlies the normal distribution in statistics). Taylor series rescue us.

PROPERTY **Term-by-term integration of a power series**

If $f(x)=\sum_{k=0}^{\infty} a_k x^k$ converges on an interval containing 0 and x, then

$$\int_0^x f(t)\,dt=\sum_{k=0}^{\infty} a_k\frac{x^{k+1}}{k+1},$$

i.e. a power series may be integrated **term by term** inside its interval of convergence, and the resulting series has the same interval of convergence. (The proof is beyond the syllabus; you may quote and use this result.)

WORKED EXAMPLE 9 **Estimating $\int_0^{0.5} e^{-x^2}\,dx$ to 4 decimal places**

Show that $\int_0^{0.5} e^{-x^2}\,dx \approx 0.4613$.

SOLUTION

Solution. Replace x by $-x^2$ in $e^x = \sum \frac{x^k}{k!}$:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \cdots$$

Integrating term by term from 0 to 0.5:

$$\int_0^{0.5} e^{-x^2}\,dx = \left[x - \frac{x^3}{3} + \frac{x^5}{5\cdot 2!} - \frac{x^7}{7\cdot 3!} + \frac{x^9}{9\cdot 4!} - \cdots\right]_0^{0.5}$$

$$= 0.5 - \frac{0.125}{3} + \frac{0.03125}{10} - \frac{0.0078125}{42} + \frac{0.001953125}{216} - \cdots$$
$$= 0.5 - 0.0416667 + 0.0031250 - 0.0001860 + 0.0000090 - \cdots$$
$$\approx 0.4612813.$$

The series is alternating with decreasing terms; the first neglected term is below $4 \times 10^{-7} < \frac{1}{2} \times 10^{-4}$, so the result is guaranteed: $\int_0^{0.5} e^{-x^2}\,dx \approx \boxed{0.4613}$ (4 d.p.).

EXEMPLE **The sine integral**

Similarly, since $\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots$ (for $x \neq 0$, with value 1 at $x = 0$),

$$\int_0^1 \frac{\sin x}{x}\,dx = 1 - \frac{1}{3\cdot 3!} + \frac{1}{5\cdot 5!} - \frac{1}{7\cdot 7!} + \cdots = 1 - 0.055556 + 0.001667 - 0.000028 + \cdots \approx 0.9461.$$

3.5 Combining with Simpson's Rule: Two Independent Estimates

We now have two ways to estimate a definite integral: Simpson's rule (Lesson 83) and term-by-term series integration. A powerful exam technique — and good scientific practice — is to compute both and check that they agree.

WORKED EXAMPLE 10 **Series versus Simpson for $\int_0^1 \frac{\sin x}{x}\,dx$**

Estimate $I = \int_0^1 \frac{\sin x}{x}\,dx$ using Simpson's rule with 4 strips, and compare with the series

value 0.9461.

SOLUTION

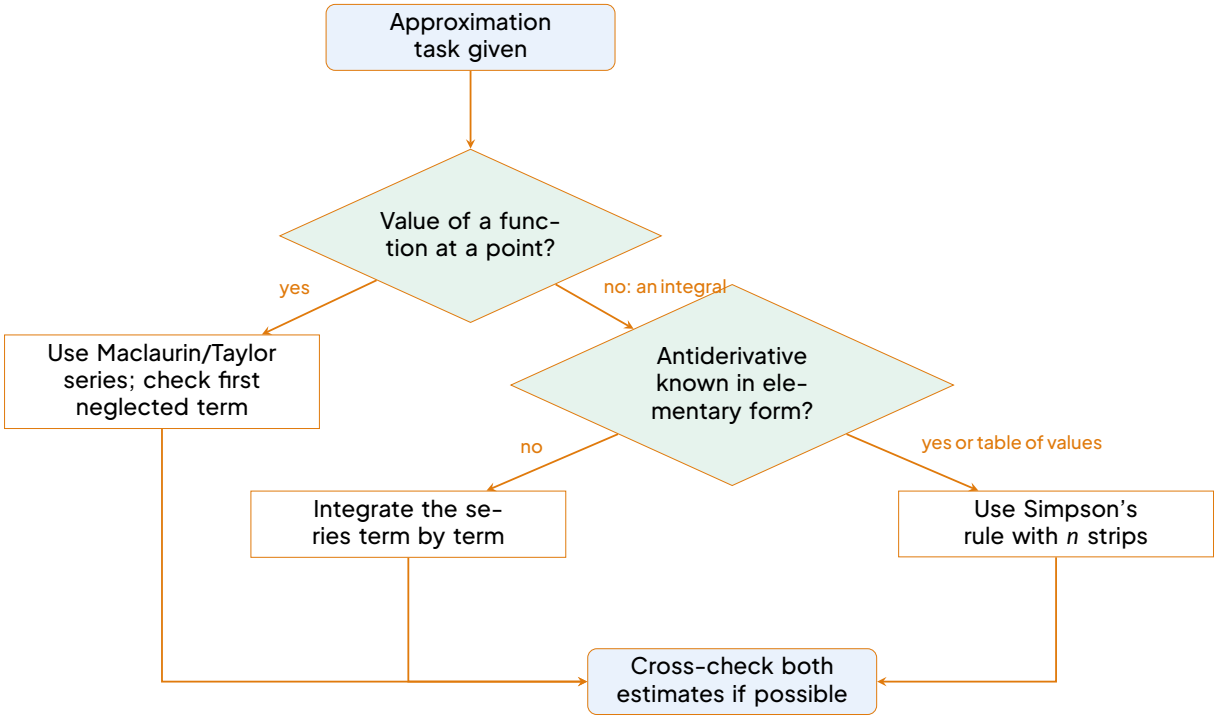
Solution. With $h = 0.25$, define $f(x) = \frac{\sin x}{x}$ with $f(0) = 1$ (the limiting value).

x	0	0.25	0.5	0.75	1
$f(x)$	1.000000	0.989616	0.958851	0.908852	0.841471

Simpson's rule:

$$I \approx \frac{h}{3}\left[f(0) + f(1) + 4\big(f(0.25) + f(0.75)\big) + 2f(0.5)\right]$$
$$= \frac{0.25}{3}\left[1 + 0.841471 + 4(0.989616 + 0.908852) + 2(0.958851)\right]$$
$$= \frac{0.25}{3}\left[1.841471 + 7.593872 + 1.917702\right] = \frac{0.25}{3}(11.353045) \approx 0.9461.$$

The two methods agree to 4 decimal places (0.94608 vs 0.94608), which strongly confirms the result.



Choosing between a Taylor series and Simpson's rule for a given approximation task.

3.6 Binomial-Type Approximations for Roots

The binomial series $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$ (valid for $|x| < 1$) is ideal for roots and reciprocals of numbers close to 1.

WORKED EXAMPLE 11 **Approximating $\sqrt{1.02}$ and $(0.98)^{-1}$**

(a) Find $\sqrt{1.02}$ to 6 decimal places. (b) Find $(0.98)^{-1}$ to 4 decimal places.

SOLUTION

Solution. (a) Write $\sqrt{1.02} = (1+0.02)^{1/2}$. With $n = \frac{1}{2}$, $x = 0.02$:

$$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots = 1 + 0.01 - \frac{0.0004}{8} + \frac{0.000008}{16} - \cdots$$

$$= 1 + 0.01 - 0.00005 + 0.0000005 - \cdots = 1.0099505$$

The next term is below 10^{-8}, so $\sqrt{1.02} \approx \boxed{1.009950}$ (6 d.p.). (b) Write $(0.98)^{-1} = (1-0.02)^{-1}$. With $n = -1$, $x = -0.02$:

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots = 1 + 0.02 + 0.0004 + 0.000008 + \cdots = 1.020408\ldots$$

The term $x^4 = 1.6 \times 10^{-7} < \frac{1}{2} \times 10^{-4}$ after summing, so $(0.98)^{-1} \approx \boxed{1.0204}$ (4 d.p.).

3.7 GCE Exam Technique

KEY POINTS **How to present a Taylor approximation in the exam**

1. State the series you use, with its general term and interval of validity.
2. Substitute the value (in radians for angles), computing each term with at least two extra decimal places.
3. Bound the error: quote the first neglected term and show it is less than half a unit of the last required decimal place.
4. Conclude with a boxed answer rounded to the requested accuracy, and where possible cross-check with another method (Simpson's rule for integrals).

EXERCISE 4 **Practice**

1. Use the Maclaurin series of $\ln(1+x)$ to find $\ln(1.1)$ correct to 5 decimal places, justifying

the number of terms.

2. Evaluate $\lim_{x\to 0}\dfrac{\sin x - x}{x^3}$ using a series.

3. Estimate $\displaystyle\int_0^{0.4}\frac{\sin x}{x}\,dx$ to 5 decimal places by term-by-term integration, then verify with Simpson's rule using 4 strips.

4. Use the binomial series to find $\sqrt[3]{1.03}$ correct to 5 decimal places.

SOLUTION **Exercise 1**

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $-1 < x \le 1$. With $x = 0.1$: $0.1 - 0.005 + 0.00033333 - 0.000025 + 0.000002 = 0.09531033\ldots$; the next term $1.7\times10^{-7} < \frac{1}{2}\times10^{-5}$. Hence $\ln(1.1) \approx 0.09531$ (5 d.p.).

SOLUTION **Exercise 2**

$\sin x - x = -\dfrac{x^3}{6} + \dfrac{x^5}{120} - \cdots$, so $\dfrac{\sin x - x}{x^3} = -\dfrac{1}{6} + \dfrac{x^2}{120} - \cdots \to -\dfrac{1}{6}$ as $x \to 0$.

SOLUTION **Exercise 3**

Series: $\left[x - \frac{x^3}{18} + \frac{x^5}{600} - \cdots\right]_0^{0.4} = 0.4 - 0.0035556 + 0.0000171 = 0.3964615$, next term $< 10^{-7}$, so $I \approx 0.39646$ (5 d.p.). Simpson with $h = 0.1$: ordinates $f(0) = 1$, $f(0.1) = 0.998334$, $f(0.2) = 0.993347$, $f(0.3) = 0.985067$, $f(0.4) = 0.973546$; $I \approx \frac{0.1}{3}[1+0.973546+4(0.998334+0.985067)+2(0.993347)] = \frac{0.1}{3}(11.893854) \approx 0.39646$. The two estimates agree.

SOLUTION **Exercise 4**

$(1.03)^{1/3} = 1 + \frac{0.03}{3} + \frac{\frac{1}{3}(-\frac{2}{3})}{2}(0.03)^2 + \frac{\frac{1}{3}(-\frac{2}{3})(-\frac{5}{3})}{6}(0.03)^3 + \cdots = 1 + 0.01 - 0.0001 + 0.00000167 - \cdots = 1.00990167\ldots$; next term $< 3\times10^{-8}$. Hence $\sqrt[3]{1.03} \approx 1.00990$ (5 d.p.).

DID YOU KNOW? **Series in your pocket**

Every scientific calculator and every computer evaluates e^x, $\sin x$, $\ln x$ by means of rapidly convergent polynomial approximations closely related to Taylor series — the very technique you have just learned. The same mathematics that computes satellite orbits and models the spread of a disease in Douala is, at its heart, a well-chosen truncated series with a controlled error.

4 STEP-BY-STEP METHODS FOR DIFFERENTIAL EQUATIONS

In the previous section we saw how Taylor series provide polynomial approximations of functions. When a function is defined implicitly as the solution of a differential equation, we can use the differential equation itself to build the approximation step by step. This is the essence of numerical methods for ordinary differential equations. They are indispensable when an exact analytical solution cannot be found — a frequent situation in real-world modelling (population dynamics, heat transfer, spread of diseases, etc.).

4.1 The Initial Value Problem and the Need for Numerical Approximation

DEFINITION **Initial Value Problem (IVP) and Step Size**

An initial value problem (IVP) for a first-order ordinary differential equation is written as

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0,$$

where f is a given function of two variables, and (x_0, y_0) is a known point on the solution curve. The step size $h > 0$ is the increment in the independent variable x used to advance from one approximation to the next. We define the mesh points $x_n = x_0 + nh$ for $n = 0, 1, 2, \ldots$

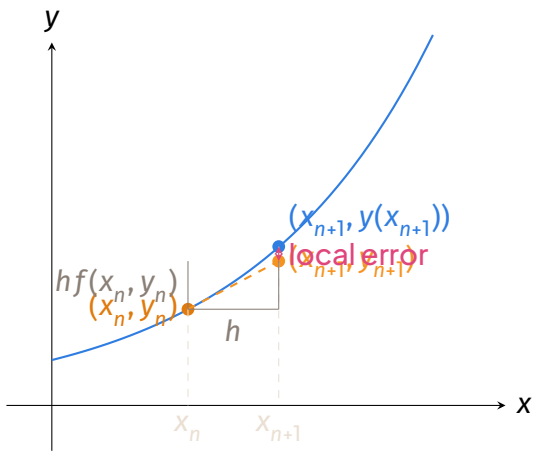
Many differential equations arising in applications — for example $\frac{dy}{dx} = e^{-x^2}$ or $\frac{dy}{dx} = \sin(x^2)$ — have no elementary antiderivative. Even when an exact solution exists, it may be too complicated to evaluate efficiently. Numerical methods produce a sequence of approximate values $y_n \approx y(x_n)$ that can be computed with simple arithmetic.

4.2 Euler's Method: The Tangent Line Approximation

Geometric idea. At the known point (x_n, y_n) the differential equation gives the slope $f(x_n, y_n)$. Over a small step h we follow the tangent line:

$$y_{n+1} = y_n + h\, f(x_n, y_n).$$

This is exactly the first-order Taylor expansion $y(x_n + h) \approx y(x_n) + h\, y'(x_n)$ with $y'(x_n) = f(x_n, y_n)$.



One step of Euler's method: the tangent segment (orange) approximates the true curve (blue). The vertical gap at x_{n+1} is the local truncation error.

METHOD **Euler's Method Algorithm**

1. Input: $f(x, y)$, initial point (x_0, y_0), step size h, number of steps N (or final x value).
2. For $n = 0, 1, \ldots, N - 1$:
   - Compute $k = f(x_n, y_n)$.
   - Update $y_{n+1} = y_n + h\,k$.
   - Update $x_{n+1} = x_n + h$.
3. Output the table of (x_n, y_n).

EXEMPLE **Euler's Method: Worked Example**

Solve $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.2$ up to $x = 1.0$.

n	x_n	y_n	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.2\,f(x_n, y_n)$
0	0.0	1.0000	1.0000	1.2000
1	0.2	1.2000	1.4000	1.4800
2	0.4	1.4800	1.8800	1.8560
3	0.6	1.8560	2.4560	2.3472
4	0.8	2.3472	3.1472	2.9766
5	1.0	2.9766	–	–

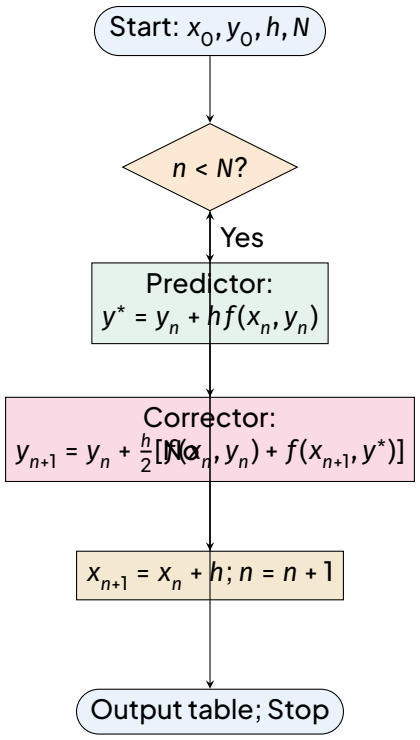
The exact solution is $y(x) = 2e^x - x - 1$; at $x = 1$ the exact value is $2e - 2 \approx 3.4366$. Euler's approximation 2.9766 underestimates by about 0.46.

4.3 Improved Euler (Heun) Method: Predictor–Corrector

Euler's method uses only the slope at the beginning of the interval. The improved Euler method (also called Heun's method) averages the slope at the start and at a predicted end point. It can be derived by approximating the integral $\int_{x_n}^{x_{n+1}} f(x,y)\,dx$ with the trapezoidal rule, using a predicted value for the endpoint.

METHOD **Improved Euler (Heun) Algorithm**

1. Input: $f(x,y)$, (x_0, y_0), h, N.
2. For $n = 0, 1, \dots, N-1$:
   - Predictor: $y^*_{n+1} = y_n + h\,f(x_n, y_n)$.
   - Corrector: $y_{n+1} = y_n + \dfrac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y^*_{n+1})\right]$.
   - $x_{n+1} = x_n + h$.
3. Output (x_n, y_n).



Flowchart of the improved Euler (Heun) predictor–corrector loop.

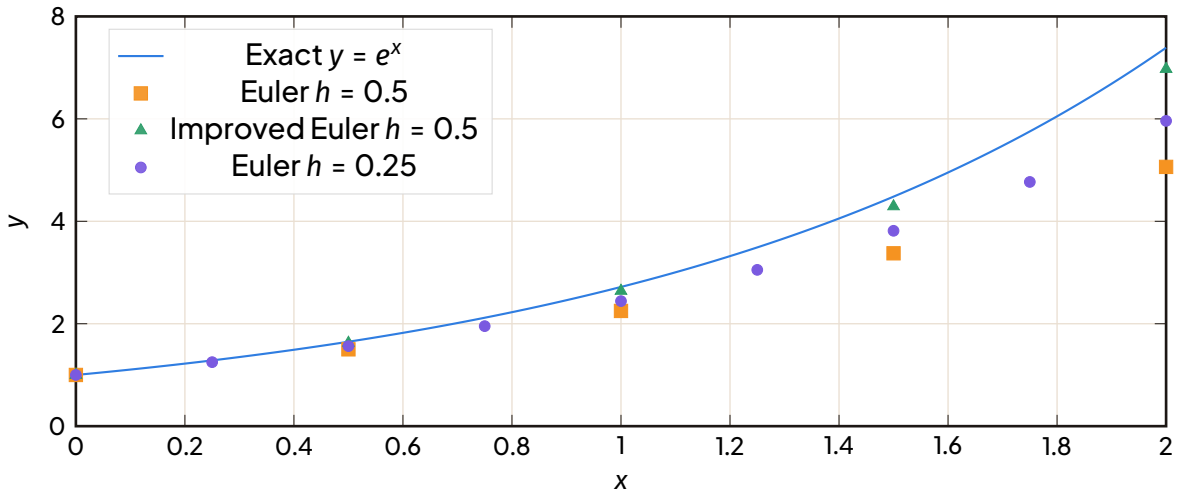
EXEMPLE **Improved Euler: Same IVP**

$\frac{dy}{dx} = x + y$, $y(0) = 1$, $h = 0.2$ up to $x = 1.0$.

n	x_n	y_n	$f(x_n, y_n)$	y^*_{n+1}	$f(x_{n+1}, y^*)$	y_{n+1}
0	0.0	1.0000	1.0000	1.2000	1.4000	1.2400
1	0.2	1.2400	1.4400	1.5280	1.9280	1.5768
2	0.4	1.5768	1.9768	1.9722	2.5722	2.0317
3	0.6	2.0317	2.6317	2.5580	3.3580	2.6307
4	0.8	2.6307	3.4307	3.3168	4.3168	3.4055
5	1.0	3.4055	–	–	–	–

The improved Euler value 3.4055 is much closer to the exact 3.4366 (error ≈ 0.031).

4.4 Comparison, Error Behaviour and Practical Considerations



Comparison of exact solution $y = e^x$ (IVP $y' = y$, $y(0) = 1$) with Euler and improved Euler for two step sizes. Smaller h reduces the error; improved Euler is consistently more accurate than Euler for the same h.

PROPERTY **Error Behaviour**

- Local truncation error (error made in one step assuming exact previous value): Euler $O(h^2)$, Improved Euler $O(h^3)$.
- Global accumulated error after N steps to reach a fixed x: Euler $O(h)$, Improved Euler $O(h^2)$.
- Halving h roughly halves the global error for Euler, but quarters it for improved Euler.
- Smaller h increases the number of steps and thus the computational cost and round-off error accumulation.

WARNING **Common Mistakes**

- Heun's second slope at the wrong point: In the corrector, the second slope must be evaluated at the predicted point (x_{n+1}, y^*_{n+1}), not at (x_n, y_n) or (x_{n+1}, y_n). Using $f(x_n, y_n)$ twice reduces the method to ordinary Euler.
- Premature rounding: Keep all intermediate values (especially y^* and the slopes) to full calculator precision. Round only the final table entries to the required number of decimal places. Rounding y^* to 4 decimals before computing the corrector can noticeably degrade accuracy.

- Confusing x_{n+1} in the corrector: Remember that $x_{n+1} = x_n + h$ is known before the corrector step; use it in $f(x_{n+1}, y^*)$.

4.5 Modelling Applications

Numerical methods shine when modelling real phenomena where the differential equation cannot be solved analytically.

EXPERIMENT **Population Growth with Limited Resources**

The logistic model $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$ describes a population $P(t)$ with growth rate r and carrying capacity K. For $r = 0.1\,\text{yr}^{-1}$, $K = 1000$, $P(0) = 100$, use improved Euler with $h = 1$ year to estimate $P(5)$.

n	t_n	P_n	$f(t_n, P_n)$	P^*_{n+1}	$f(t_{n+1}, P^*)$	P_{n+1}
0	0	100.00	9.00	109.00	9.71	109.36
1	1	109.36	9.73	119.09	10.48	119.47
2	2	119.47	10.49	129.96	11.21	130.32
3	3	130.32	11.22	141.54	11.89	141.88
4	4	141.88	11.89	153.77	12.50	154.08
5	5	154.08	–	–	–	–

The population grows from 100 to about 154 in five years. The table shows the growth rate increasing as the population approaches half the carrying capacity.

DID YOU KNOW? **Newton's Law of Cooling in a Cameroonian Context**

A pot of hot ndolé (initial temperature 95°C) cools in a kitchen at 28°C. The temperature $T(t)$ satisfies $\frac{dT}{dt} = -k(T - 28)$. If $k = 0.07\ \text{min}^{-1}$, Euler's method with $h = 2$ min gives a quick estimate of when the stew reaches a safe serving temperature (60°C). Such step-by-step calculations are easily programmed on a smartphone — a practical tool for food safety in street-food vendors' stalls.

4.6 Worked Examination-Style Question

WORKED EXAMPLE 12 **Combined Euler and Improved Euler**

Consider the IVP $\frac{dy}{dx} = y - x$, $y(0) = 2$.

1. Use Euler's method with $h = 0.2$ to approximate $y(0.6)$.
2. Use the improved Euler method with the same h to approximate $y(0.6)$.
3. The exact solution is $y(x) = x + 1 + e^x$. Compute the exact value $y(0.6)$ and the absolute errors of both methods.

SOLUTION

Solution.

1. Euler's method.

n	x_n	y_n	$f(x_n, y_n) = y_n - x_n$	$y_{n+1} = y_n + 0.2\, f(x_n, y_n)$
0	0.0	2.0000	2.0000	2.4000
1	0.2	2.4000	2.2000	2.8400
2	0.4	2.8400	2.4400	3.3280
3	0.6	3.3280	–	–

Euler approximation: $y(0.6) \approx 3.3280$.

2. Improved Euler method.

n	x_n	y_n	$f(x_n, y_n)$	y^*_{n+1}	$f(x_{n+1}, y^*)$	y_{n+1}
0	0.0	2.0000	2.0000	2.4000	2.2000	2.4200
1	0.2	2.4200	2.2200	2.8640	2.4640	2.8888
2	0.4	2.8888	2.4888	3.3866	2.7866	3.4163
3	0.6	3.4163	–	–	–	–

Improved Euler approximation: $y(0.6) \approx 3.4163$.

3. Exact value and errors. $y(0.6) = 0.6 + 1 + e^{0.6} = 1.6 + 1.8221188 = 3.4221188$.

- Euler error: $|3.3280 - 3.4221| = 0.0941$.
- Improved Euler error: $|3.4163 - 3.4221| = 0.0058$.

Improved Euler is about 16 times more accurate for the same step size.

4.7 Consolidation Exercises

EXERCISE 5 **Exercise 1**

Use Euler's method with $h = 0.1$ to approximate $y(0.3)$ for $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$. Present your work in a table.

SOLUTION **Exercise 1**

n	x_n	y_n	$f(x_n, y_n) = x_n^2 + y_n^2$	y_{n+1}
0	0.0	1.0000	1.0000	1.1000
1	0.1	1.1000	1.2200	1.2220
2	0.2	1.2220	1.5333	1.3753
3	0.3	1.3753	–	–

$y(0.3) \approx 1.3753$.

EXERCISE 6 **Exercise 2**

For the IVP $\frac{dy}{dx} = 2xy$, $y(0) = 1$, the exact solution is $y = e^{x^2}$.

1. Apply the improved Euler method with $h = 0.2$ to estimate $y(0.6)$.
2. Calculate the percentage error relative to the exact value.

SOLUTION **Exercise 2**

n	x_n	y_n	$f = 2x_n y_n$	y^*	$f(x_{n+1}, y^*)$	y_{n+1}
0	0.0	1.0000	0.0000	1.0000	0.4000	1.0400
1	0.2	1.0400	0.4160	1.1232	0.8986	1.1715
2	0.4	1.1715	0.9372	1.3589	1.6307	1.4282
3	0.6	1.4282	–	–	–	–

Exact $y(0.6) = e^{0.36} = 1.4333$. Percentage error $= \frac{|1.4282 - 1.4333|}{1.4333} \times 100\% \approx 0.36\%$.

KEY POINTS **Key Points for the Examination**

- Euler: $y_{n+1} = y_n + hf(x_n, y_n)$; global error $O(h)$.
- Improved Euler (Heun): predictor $y^* = y_n + hf(x_n, y_n)$, corrector $y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, y^*)]$; global error $O(h^2)$.
- Always organise calculations in a clear table showing x_n, y_n, slopes, predictor, corrector.
- Do not round intermediate values; round only the final presented answers.
- In Heun's corrector, the second slope uses the predicted y^* at x_{n+1}.
- Numerical methods are essential for modelling real-world problems where analytical solutions are unavailable.

5 INTEGRATION ACTIVITY

6 INTEGRATION ACTIVITY: SURVEYING THE WOURI RIVER CROSSING

6.1 Aims of the Activity

This integration activity brings together the three numerical techniques of this chapter in a single, authentic engineering task. By the end of the activity, you should be able to:

- apply **Simpson's rule** to real discrete data (equally spaced depth soundings) to estimate a cross-sectional area and a derived physical quantity (discharge);
- construct a **Maclaurin polynomial** of an exponential function, integrate it term by term, and assess its accuracy against a Simpson estimate and the exact value;
- implement **Euler's method** and the **improved Euler method** (Heun's method) for a first-order differential equation, and compare both with the exact solution;
- organise numerical work in clear tables, control rounding errors, and argue critically about the reliability and efficiency of a numerical method.

KEY POINTS **Skills Mobilised**

Numerical integration (Simpson's rule) • Power series approximation (Taylor/Maclaurin) • Step-by-step solution of $\frac{dy}{dx} = f(x, y)$ (Euler, Improved Euler) • Error awareness (truncation, rounding) • Scientific communication and teamwork.

6.2 Situation: Surveying the Wouri River Crossing

Context.

A civil engineering team is preparing a feasibility study for a water-intake station on the **Wouri River**, near Douala. Before any construction, the team must characterise a straight cross-section of the river at the chosen site. Three questions arise:

1. What is the discharge (volume of water flowing per second) through the section?
2. What is the mean current speed over the width?
3. How does a pollutant or sediment concentration evolve in time downstream?

Data Sheet 1 — Depth Soundings.

Using an echo-sounder from a boat, technicians measured the depth d (in metres) every 4 m along a 32 m-wide section, from the left bank ($x = 0$) to the right bank ($x = 32$):

x (m)	0	4	8	12	16	20	24	28	32
d (m)	0	1.8	3.2	4.5	5.1	4.8	3.6	2.1	0

The surface current speed, measured with a float over a timed distance, is $v_s = 0.8\ \text{m s}^{-1}$; the team assumes this value is uniform across the section for a first discharge estimate.

Data Sheet 2 — Velocity Profile.

A more refined model gives the current speed at distance x from the left bank as

$$v(x) = 1.5\,e^{-0.05x}\ (\text{in m s}^{-1}), \qquad 0 \le x \le 32.$$

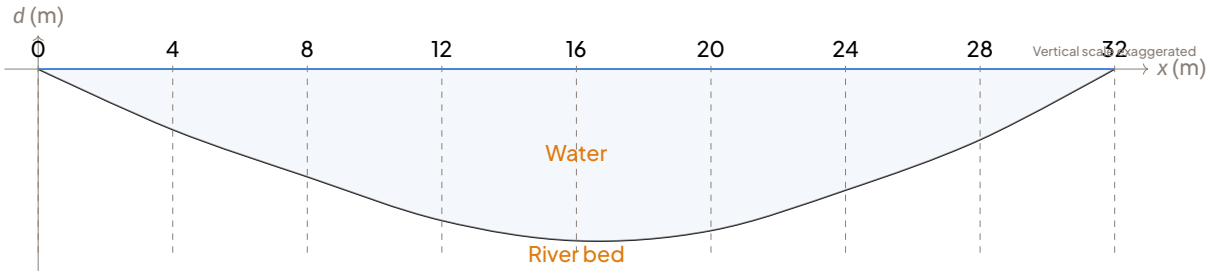
The **mean speed** across the section is defined by $\bar{v} = \dfrac{1}{32}\displaystyle\int_0^{32} v(x)\,dx$.

Data Sheet 3 — Sediment Decay.

Sediment concentration C (in mg/L) in the plume downstream satisfies

$$\frac{dC}{dt} = -0.05\,C, \qquad C(0) = 80\ \text{mg/L},$$

where t is the time in hours. The team wants an estimate of $C(2)$.



Cross-section of the Wouri River from the depth soundings (vertical scale exaggerated for clarity).

6.3 Tasks

Work in groups of three or four. **Keep at least four decimal places in all intermediate computations.**

Task 1 — Cross-sectional Area and Discharge (Simpson's Rule).

1. State Simpson's rule for n strips of width h and verify that the data sheet is compatible with it (number of ordinates, equal spacing).
2. Estimate the cross-sectional area A of the river section.
3. Deduce the discharge (volume of water flowing per second) $Q = A\,v_s$, in $\text{m}^3\ \text{s}^{-1}$.

Task 2 — Mean Current Speed (Taylor Polynomial versus Simpson).

1. Write the Maclaurin series of e^{u}, then substitute $u = -0.05x$ to obtain the Taylor polynomial $T_4(x)$ of degree 4 of $e^{-0.05x}$.

2. Compute $\int_0^{32} 1.5\,T_4(x)\,dx$ term by term, and deduce an approximation $\bar{v}_T$ of the mean speed.
3. Evaluate $v(x)$ at the nine sounding positions and apply Simpson's rule to obtain a second approximation $\bar{v}_S$.
4. Compute the exact value $\bar{v} = \frac{1}{32}\int_0^{32} 1.5e^{-0.05x}\,dx$ and compare the three results. Comment on the accuracy of each method.

Task 3 — Sediment Concentration (Step-by-Step Methods).

1. Using Euler's method with step $h = 0.5$ h, build the table of values (t_n, C_n) up to $t = 2$ and estimate $C(2)$.
2. Repeat with the improved Euler method (Heun's method): $C^* = C_n + h\,f(C_n)$, then $C_{n+1} = C_n + \frac{h}{2}\big[f(C_n) + f(C^*)\big]$.
3. Solve the differential equation exactly and compute $C(2)$. Compare the two numerical estimates with the exact value. Calculate the absolute errors.

Task 4 — Synthesis.

Prepare a five-minute presentation: show your tables, state which method you trust most in each task, and justify your choice in terms of accuracy, computational cost (number of function evaluations), and suitability for the type of data (discrete vs. continuous function).

6.4 Model Answer

Task 1.1 — Simpson's Rule Statement and Compatibility Check.

For a function $y(x)$ tabulated at $n+1$ equally spaced ordinates $x_0, x_1, \dots, x_n$ with $x_k = a + kh$, $h = \frac{b-a}{n}$, and n **even**, Simpson's rule states:

$$\int_a^b y(x)\,dx \approx \frac{h}{3}\Big[y_0 + y_n + 4\sum_{\text{odd } k} y_k + 2\sum_{\text{even } k,\,k\neq 0,n} y_k\Big].$$

Check: Here $a = 0$, $b = 32$, $h = 4$. Number of strips $n = \frac{32-0}{4} = 8$ (even). Number of ordinates = 9. The data is perfectly compatible.

Task 1.2 — Cross-Sectional Area A.

Depths d_k: $d_0 = 0$, $d_1 = 1.8$, $d_2 = 3.2$, $d_3 = 4.5$, $d_4 = 5.1$, $d_5 = 4.8$, $d_6 = 3.6$, $d_7 = 2.1$, $d_8 = 0$.

$$\begin{aligned} A &\approx \frac{4}{3}\Big[d_0 + d_8 + 4(d_1 + d_3 + d_5 + d_7) + 2(d_2 + d_4 + d_6)\Big] \\ &= \frac{4}{3}\Big[0 + 0 + 4(1.8 + 4.5 + 4.8 + 2.1) + 2(3.2 + 5.1 + 3.6)\Big] \\ &= \frac{4}{3}\Big[4(13.2) + 2(11.9)\Big] = \frac{4}{3}\big[52.8 + 23.8\big] = \frac{4}{3}(76.6) \\ &= \frac{306.4}{3} \approx 102.1333\ \text{m}^2. \end{aligned}$$

Task 1.3 — Discharge Q.

$$Q = A \times v_s \approx 102.1333 \times 0.8 = 81.7067\ \text{m}^3\,\text{s}^{-1}.$$

Rounded to one decimal: $Q \approx 81.7\ \text{m}^3\,\text{s}^{-1}$.

Task 2.1 — Maclaurin Polynomial $T_4(x)$ for $e^{-0.05x}$.

The Maclaurin series for e^u is $e^u = \sum_{k=0}^{\infty} \frac{u^k}{k!} = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \cdots$. Substitute $u = -0.05x$:

$$\begin{aligned} e^{-0.05x} \approx T_4(x) &= 1 + (-0.05x) + \frac{(-0.05x)^2}{2} + \frac{(-0.05x)^3}{6} + \frac{(-0.05x)^4}{24} \\ &= 1 - 0.05x + \frac{0.0025}{2}x^2 - \frac{0.000125}{6}x^3 + \frac{0.00000625}{24}x^4 \\ &= 1 - 0.05x + 0.00125x^2 - 0.0000208333x^3 + 0.000000260417x^4. \end{aligned}$$

Task 2.2 — Mean Speed via Taylor Polynomial $\bar{v}_T$.

Integrate 1.5 $T_4(x)$ term by term from 0 to 32:

$$\begin{aligned} \int_0^{32} T_4(x)\,dx &= \left[x - 0.025x^2 + \frac{0.00125}{3}x^3 - \frac{0.0000208333}{4}x^4 + \frac{0.000000260417}{5}x^5\right]_0^{32} \\ &= 32 - 0.025(1024) + \frac{0.00125}{3}(32768) - \frac{0.0000208333}{4}(1048576) + \frac{0.000000260417}{5}(335 \\ &= 32 - 25.6 + 13.65333 - 5.46133 + 1.74763 \\ &= 16.33963. \end{aligned}$$

$$\bar{v}_T = \frac{1.5 \times 16.33963}{32} = \frac{24.50945}{32} \approx 0.76592\ \text{m}\,\text{s}^{-1}.$$

Task 2.3 — Mean Speed via Simpson's Rule $\bar{v}_S$.

Evaluate $v(x) = 1.5e^{-0.05x}$ at the nine points (keeping 4 decimals):

x (m)	0	4	8	12	16	20	24	28	32
$v(x)$ (m/s)	1.5000	1.2281	1.0055	0.8232	0.6740	0.5518	0.4518	0.3699	0.3029

Apply Simpson's rule ($h = 4$, $n = 8$):

$$\begin{aligned} \int_0^{32} v(x)\,dx &\approx \frac{4}{3}\left[v_0 + v_8 + 4(v_1 + v_3 + v_5 + v_7) + 2(v_2 + v_4 + v_6)\right] \\ &= \frac{4}{3}\left[1.5000 + 0.3029 + 4(1.2281 + 0.8232 + 0.5518 + 0.3699) + 2(1.0055 + 0.6740 + 0.4518)\right] \\ &= \frac{4}{3}\left[1.8029 + 4(2.9730) + 2(2.1313)\right] \\ &= \frac{4}{3}\left[1.8029 + 11.8920 + 4.2626\right] = \frac{4}{3}(17.9575) = 23.94333. \end{aligned}$$

$$\bar{v}_S = \frac{23.94333}{32} \approx 0.74823\ \text{m}\,\text{s}^{-1}.$$

Task 2.4 — Exact Value and Comparison.

Exact integral:

$$\int_0^{32} 1.5e^{-0.05x}\,dx = \left[\frac{1.5}{-0.05}e^{-0.05x}\right]_0^{32} = -30\left(e^{-1.6} - 1\right) = 30(1 - e^{-1.6}).$$

$e^{-1.6} \approx 0.2018965$.

$$\int_0^{32} v(x)\,dx = 30 \times 0.7981035 = 23.943105.$$

$$\bar{v}_{\text{exact}} = \frac{23.943105}{32} = 0.748222\ \text{m s}^{-1}.$$

Method	Mean Speed $\bar{v}$ (m s^{-1})	Absolute Error	Relative Error
Taylor (T_4)	0.76592	+0.01770	+2.37%
Simpson	0.74823	+0.00001	+0.001%
Exact	0.74822	-	-

Comment: Simpson's rule, applied to the exact function values, is exact for polynomials up to degree 3 and extremely accurate for smooth functions like $e^{-0.05x}$; it matches the exact value to 5 decimal places. The degree-4 Maclaurin polynomial is accurate near $x = 0$ but its error grows with $|x|$; at $x = 32$, $u = -1.6$ is far from the expansion centre, causing a 2.4% overestimate. Increasing the polynomial degree would improve accuracy, but Simpson's rule is far more efficient for this definite integral.

Task 3.1 — Euler's Method ($h = 0.5$ h).

$\frac{dC}{dt} = f(C) = -0.05C$, $C_0 = 80$. Euler iteration: $C_{n+1} = C_n + hf(C_n) = C_n - 0.025C_n = 0.975\,C_n$.

t_n (h)	0	0.5	1.0	1.5	2.0
Euler C_n (mg/L)	80.0000	78.0000	76.0500	74.1488	72.2950

Euler estimate: $C(2) \approx 72.2950$ mg/L.

Task 3.2 — Improved Euler Method (Heun's Method).

Predictor: $C^* = C_n + hf(C_n) = 0.975\,C_n$. Corrector: $C_{n+1} = C_n + \frac{h}{2}\left[f(C_n) + f(C^*)\right] = C_n + 0.25\left[-0.05C_n - 0.05C^*\right] = C_n - 0.0125(C_n + C^*)$. Substitute C^*: $C_{n+1} = C_n - 0.0125(1.975C_n) = (1 - 0.0246875)C_n = 0.9753125\,C_n$.

t_n (h)	0	0.5	1.0	1.5	2.0
Improved Euler C_n (mg/L)	80.0000	78.0250	76.0993	74.2214	72.3895

Improved Euler estimate: $C(2) \approx 72.3895$ mg/L.

Task 3.3 — Exact Solution and Comparison.

The ODE $\frac{dC}{dt} = -0.05C$ with $C(0) = 80$ solves to $C(t) = 80e^{-0.05t}$.

$$C(2) = 80e^{-0.1} = 80 \times 0.904837418 \approx 72.38699\ \text{mg/L}.$$

Method	$C(2)$ (mg/L)	Absolute Error	Relative Error
Euler	72.2950	-0.0920	-0.127%
Improved Euler	72.3895	+0.0025	+0.0035%
Exact	72.3870	-	-

Analysis: Improved Euler is roughly 40 times more accurate than basic Euler for the same step size ($h = 0.5$), at the cost of two evaluations of f per step instead of one. Both methods underestimate the decay (since $f(C) = -0.05C$ is concave up, Euler underestimates the drop; Improved Euler corrects this). Halving the step size would reduce Euler's error by ≈ 2 and Improved Euler's by ≈ 4.

6.5 Synthesis — Expected Conclusions

- **Discrete Data (Task 1):** **Simpson's rule** is the natural, optimal tool. It uses only the measured ordinates, requires no model fitting, and is exact for polynomials up to degree 3. It is the standard in hydrology for discharge estimation from soundings.
- **Continuous Function (Task 2):** A **Taylor/Maclaurin polynomial** provides an analytic approximation, excellent near the expansion centre ($x = 0$), but its accuracy deteriorates over a wide interval ($x \in [0, 32]$) unless the degree is high. **Simpson's rule** applied directly to the function values is vastly more reliable and efficient for definite integrals of known smooth functions.
- **Differential Equations (Task 3):** The **improved Euler method** (order 2) dominates basic Euler (order 1) at equal step size. For a given accuracy, Improved Euler allows a larger step size, reducing total function evaluations. In professional codes, higher-order Runge-Kutta methods (e.g., RK4) are standard.

WARNING **Common Mistakes to Avoid**

- Using Simpson's rule with an odd number of strips (requires even n).
- Forgetting the weight pattern $1, 4, 2, 4, \ldots, 2, 4, 1$ (first and last weight 1, odd indices 4, even indices 2).
- Mixing units (e.g., hours vs seconds in Task 3; discharge in m^3/s requires speed in m/s).
- Rounding intermediate values too early (keep 4+ decimals, round only final answer).
- Assuming a Taylor polynomial is uniformly accurate on a large interval far from the centre of expansion.
- In Improved Euler, forgetting to use the predictor C^* inside the corrector formula.

DID YOU KNOW?

Why This Matters in Cameroon

Discharge estimates like those of Task 1 are exactly what hydrologists produce on the Wouri, the Sanaga, the Benue, or the Logone before sizing bridges, water intake stations, irrigation canals, or small hydropower plants. The same numerical methods you have just used by hand — Simpson's rule for integration, Taylor expansions for modelling, Runge-Kutta schemes for differential equations — are the computational engines of the professional software (HEC-RAS, Mike 11, SWAT) used by the Ministry of Water and Energy and consulting firms. The only difference is the number of data points: field campaigns may yield hundreds of soundings, and time-stepping runs for thousands of steps, but the mathematical principles remain identical.

7 METHODS AND WORKED EXAMPLES

7.1 Applying Simpson's Rule with a Given Number of Strips

METHOD **Using Simpson's Rule**

To estimate $I = \int_a^b f(x)\,dx$ using Simpson's rule with n strips (n even):

1. Compute the strip width $h = \dfrac{b-a}{n}$.
2. List the ordinates $y_0, y_1, \ldots, y_n$ where $y_r = f(a + rh)$, working to at least one decimal place more than required in the final answer.
3. Apply the formula

$$I \approx \frac{h}{3}\left[y_0 + y_n + 4\left(y_1 + y_3 + \cdots + y_{n-1}\right) + 2\left(y_2 + y_4 + \cdots + y_{n-2}\right)\right].$$

4. Pattern to remember: **first + last**, plus **4 times the odd** ordinates, plus **2 times the even** ordinates (excluding the end ones).
5. Round only at the very end; give the answer to the accuracy requested.

Reflex: check that n is even before starting; if the question says "5 ordinates", that means $n = 4$ strips.

WORKED EXAMPLE 13 **Simpson's rule with 4 strips**

Use Simpson's rule with 4 strips to estimate $\int_0^1 \dfrac{1}{1+x^2}\,dx$, giving your answer to 4 decimal places.

SOLUTION

Solution. Here $a = 0$, $b = 1$, $n = 4$, so $h = \dfrac{1-0}{4} = 0.25$. The ordinates are $y_r = f(0.25r)$ with

$f(x) = \dfrac{1}{1+x^2}$:

$$y_0 = f(0) = 1.000000,$$
$$y_1 = f(0.25) = \frac{1}{1.0625} = 0.941176,$$
$$y_2 = f(0.5) = \frac{1}{1.25} = 0.800000,$$
$$y_3 = f(0.75) = \frac{1}{1.5625} = 0.640000,$$
$$y_4 = f(1) = 0.500000.$$

Applying Simpson's rule:

$$I \approx \frac{0.25}{3}\left[y_0 + y_4 + 4(y_1 + y_3) + 2y_2\right]$$
$$= \frac{0.25}{3}\left[1.000000 + 0.500000 + 4(0.941176 + 0.640000) + 2(0.800000)\right]$$
$$= \frac{0.25}{3}\left[1.5 + 4(1.581176) + 1.6\right]$$
$$= \frac{0.25}{3}\left[9.424706\right] = 0.785392.$$

Hence $\displaystyle\int_0^1 \frac{dx}{1+x^2} \approx 0.7854$ (4 d.p.). Remark: the exact value is $\tan^{-1}1 = \dfrac{\pi}{4} = 0.785398\ldots$ — Simpson's rule with only 4 strips already gives 5 correct decimal places.

7.2 Simpson's Rule from Tabulated Data

METHOD **Simpson's Rule with Experimental Data**

When the function is given only as a **table of values** (as in an experiment):

1. Read the strip width h from the table (the common difference of the x-values). Check the values are equally spaced and that the number of strips is even.
2. Identify the ordinates $y_0, \ldots, y_n$ directly from the table — no evaluation of f is needed.
3. Apply Simpson's formula exactly as before.
4. If the integral has a physical meaning (area, distance from a velocity–time table, work done), state the units in the answer.

Reflex: the data points are the ordinates; do not try to find a formula for f.

WORKED EXAMPLE 14 **Estimating a distance travelled**

The speed v (in m/s) of a car travelling between Buea and Mutengene is recorded every 10 seconds:

t (s)	0	10	20	30	40	50	60
v (m/s)	0	12.5	18.2	20.1	17.6	13.0	8.4

Use Simpson's rule to estimate the distance travelled during these 60 seconds, giving your answer to the nearest metre.

SOLUTION

Solution. The distance is $s = \int_0^{60} v\,dt$. The readings are equally spaced with $h = 10$ and there are $n = 6$ strips (even, so Simpson applies). The ordinates are

$$y_0 = 0,\ y_1 = 12.5,\ y_2 = 18.2,\ y_3 = 20.1,\ y_4 = 17.6,\ y_5 = 13.0,\ y_6 = 8.4.$$

Simpson's rule:

$$\begin{aligned} s &\approx \frac{10}{3}\left[y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)\right] \\ &= \frac{10}{3}\left[0 + 8.4 + 4(12.5 + 20.1 + 13.0) + 2(18.2 + 17.6)\right] \\ &= \frac{10}{3}\left[8.4 + 4(45.6) + 2(35.8)\right] \\ &= \frac{10}{3}\left[8.4 + 182.4 + 71.6\right] = \frac{10}{3}(262.4) = 874.67. \end{aligned}$$

The car travelled approximately $\boxed{875\text{ m}}$ (to the nearest metre).

7.3 Comparing Estimates and Assessing Accuracy

METHOD **Improving and Checking a Simpson Estimate**

1. Compute the estimate S_n with n strips, then S_{2n} with $2n$ strips (halving h). Reuse the old ordinates: the new ordinates are the midpoints.
2. If S_n and S_{2n} agree to k decimal places, the estimate is likely correct to that accuracy.
3. Remember: Simpson's rule is exact for polynomials of degree up to 3, because its error term involves $f^{(4)}$.
4. For a function whose fourth derivative is small on $[a, b]$, Simpson's rule is very accurate even for modest n.

Reflex: if a question asks to show an integral is approximately a given value, compute with two different strip numbers and compare.

WORKED EXAMPLE 15 **Two strip numbers, then comparison**

(a) Use Simpson's rule with 2 strips and then with 4 strips to estimate $I = \int_1^2 \ln x \, dx$, giving each answer to 4 decimal places.
(b) By evaluating I exactly, comment on the accuracy of the 4–strip estimate.

SOLUTION

Solution. (a) With $n = 2$: $h = \frac{2-1}{2} = 0.5$; ordinates at $x = 1, 1.5, 2$:

$$y_0 = \ln 1 = 0, \quad y_1 = \ln 1.5 = 0.405465, \quad y_2 = \ln 2 = 0.693147.$$

$$S_2 = \frac{0.5}{3}\left[0 + 0.693147 + 4(0.405465)\right] = \frac{0.5}{3}(2.315007) = 0.3858 \text{ (4 d.p.)}.$$

With $n = 4$: $h = 0.25$; new ordinates at $x = 1.25$ and 1.75:

$$\ln 1.25 = 0.223144, \qquad \ln 1.75 = 0.559616.$$

$$\begin{aligned} S_4 &= \frac{0.25}{3}\left[0 + 0.693147 + 4(0.405465 + 0.559616) + 2(0.223144)\right] \\ &= \frac{0.25}{3}\left[0.693147 + 4(0.965081) + 0.446288\right] \\ &= \frac{0.25}{3}(4.999759) = 0.3863 \text{ (4 d.p.)}. \end{aligned}$$

(b) Exactly:

$$I = \left[x \ln x - x\right]_1^2 = (2\ln 2 - 2) - (0 - 1) = 2\ln 2 - 1 = 0.386294.$$

The 4–strip estimate 0.3863 agrees with the exact value to 4 decimal places — an error of less than 4×10^{-6}. Halving h from S_2 to S_4 reduced the error dramatically, illustrating the high accuracy of Simpson's rule.

7.4 Deriving Taylor and Maclaurin Polynomials

METHOD **Finding a Maclaurin Polynomial**

To find the Maclaurin polynomial of degree n for $f(x)$:

1. Compute the derivatives $f'(x), f''(x), \ldots, f^{(n)}(x)$.

2. Evaluate each at $x = 0$.
3. Substitute into
$$f(x) \approx f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0).$$
4. For a Taylor expansion about $x = a$, replace x by $(x - a)$ and evaluate at a:
$$f(x) \approx f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \cdots$$

Reflexes: learn the standard series — e^x, $\sin x$, $\cos x$, $\ln(1 + x)$, $(1 + x)^n$ — and build others by substitution, e.g. replace x by $2x$ or by $-x^2$.

WORKED EXAMPLE 16 **Maclaurin series of $e^{-x}\sin x$**

Find the Maclaurin expansion of $f(x) = e^{-x}\sin x$ up to and including the term in x^4.

SOLUTION

Solution. Method 1: direct differentiation.

$$\begin{aligned} f(x) &= e^{-x}\sin x & f(0) &= 0 \\ f'(x) &= e^{-x}(\cos x - \sin x) & f'(0) &= 1 \\ f''(x) &= -2e^{-x}\cos x & f''(0) &= -2 \\ f'''(x) &= 2e^{-x}(\cos x + \sin x) & f'''(0) &= 2 \\ f^{(4)}(x) &= -4e^{-x}\sin x & f^{(4)}(0) &= 0 \end{aligned}$$

Therefore
$$f(x) \approx 0 + x + \frac{-2}{2!}x^2 + \frac{2}{3!}x^3 + \frac{0}{4!}x^4 = x - x^2 + \frac{x^3}{3}.$$

Method 2 (check): multiplying standard series.

$$\begin{aligned} e^{-x}\sin x &= \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots\right)\left(x - \frac{x^3}{6} + \cdots\right) \\ &= x - x^2 + \frac{x^3}{2} - \frac{x^4}{6} - \frac{x^3}{6} + \frac{x^4}{6} + \cdots \\ &= x - x^2 + \frac{x^3}{3} + 0 \cdot x^4 + \cdots \end{aligned}$$

Both methods agree:
$$e^{-x}\sin x \approx x - x^2 + \frac{x^3}{3} \quad \text{(up to } x^4\text{)}.$$

7.5 Using Taylor Approximations for Numerical Values and Limits

METHOD **Numerical Approximation via Series**

1. Identify a suitable standard series and the value of x to substitute (choose x small so convergence is fast).
2. Substitute, keeping enough terms for the required accuracy.
3. For limits of indeterminate form $\frac{0}{0}$: replace each function by its Maclaurin polynomial, cancel the common power of x, then let $x \to 0$.
4. For small-angle work, remember $\sin x \approx x$, $\cos x \approx 1 - \frac{x^2}{2}$, $\tan x \approx x$ (x in radians).

Reflex: the first neglected term indicates the size of the error.

WORKED EXAMPLE 17 **Series for a value and a limit**

(a) Use the Maclaurin series of $\cos x$ up to the term in x^4 to estimate $\cos(0.3)$, giving 6 decimal places.

(b) Using Maclaurin series, evaluate $\lim_{x\to 0} \dfrac{e^x - 1 - x}{1 - \cos x}$.

SOLUTION

Solution. (a) $\cos x \approx 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!}$. With $x = 0.3$:

$$\cos(0.3) \approx 1 - \frac{0.09}{2} + \frac{0.0081}{24} = 1 - 0.045 + 0.0003375 = 0.955338.$$

(The calculator value is 0.955336; the neglected term $\frac{x^6}{720} \approx 10^{-6}$ explains the tiny difference.)

(b) Use $e^x = 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{6} + \cdots$ and $\cos x = 1 - \dfrac{x^2}{2} + \dfrac{x^4}{24} - \cdots$:

$$\frac{e^x - 1 - x}{1 - \cos x} = \frac{\frac{x^2}{2} + \frac{x^3}{6} + \cdots}{\frac{x^2}{2} - \frac{x^4}{24} + \cdots} = \frac{\frac{1}{2} + \frac{x}{6} + \cdots}{\frac{1}{2} - \frac{x^2}{24} + \cdots}.$$

As $x \to 0$, this tends to $\dfrac{1/2}{1/2} = 1$. Hence $\lim_{x\to 0} \dfrac{e^x - 1 - x}{1 - \cos x} = 1$.

7.6 Euler's Step-by-Step Method

METHOD **Euler's Method for $y' = f(x, y)$**

To solve $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$ numerically, with step h:

1. Write the recurrence (the **Euler formula**):

$$y_{n+1} = y_n + h\, f(x_n, y_n), \qquad x_{n+1} = x_n + h.$$

2. Set out a table with columns n, x_n, y_n, $f(x_n, y_n)$, $h\, f(x_n, y_n)$.
3. Compute row by row, keeping full calculator precision until the end.
4. The gradient used at each step is the gradient at the beginning of the step.

Reflex: $y_{n+1} = y_n + h \times$ (gradient at current point) — "new = old + step × slope".

WORKED EXAMPLE 18 **Euler's method**

Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with step $h = 0.1$ to estimate $y(0.3)$, giving your answer to 4 decimal places.

SOLUTION

Solution. Here $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$. Step 1: $f(0, 1) = 0 + 1 = 1$, so

$$y_1 = 1 + 0.1(1) = 1.1.$$

Step 2: $f(0.1, 1.1) = 0.1 + 1.1 = 1.2$, so

$$y_2 = 1.1 + 0.1(1.2) = 1.22.$$

Step 3: $f(0.2, 1.22) = 0.2 + 1.22 = 1.42$, so

$$y_3 = 1.22 + 0.1(1.42) = 1.362.$$

Hence $y(0.3) \approx 1.3620$ (4 d.p.). Check: the exact solution of $y' = x + y$, $y(0) = 1$ is $y = 2e^x - x - 1$, giving $y(0.3) = 2e^{0.3} - 1.3 = 1.3997$. Euler's estimate is below the true value — the true curve is convex, and Euler follows tangent lines, which lie below the curve.

7.7 The Improved Euler (Heun) Method

METHOD **Improved Euler / Heun's Method**

For $y' = f(x, y)$, $y(x_0) = y_0$, step h:

1. Compute the predictor (an ordinary Euler step):

$$\tilde{y}_{n+1} = y_n + h\, f(x_n, y_n).$$

2. Evaluate the gradient at the predicted point: $f(x_{n+1}, \tilde{y}_{n+1})$.
3. Compute the corrector using the average of the two gradients:

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1})\right].$$

4. Repeat for each step. This is a second-order method: the error behaves like h^2, much better than Euler's h.

Reflex: predict with Euler, then correct by averaging the slope at the start and the slope at the predicted end.

WORKED EXAMPLE 19 **Improved Euler method**

Given $\dfrac{dy}{dx} = x + y$ with $y(0) = 1$, use the improved Euler method with step $h = 0.1$ to estimate $y(0.2)$, giving your answer to 4 decimal places. Compare with the exact value $y(0.2) = 1.2428$.

SOLUTION

Solution. $f(x, y) = x + y$, $h = 0.1$. Step 1 (from $(0, 1)$):

$$\begin{aligned} \text{Predictor:}\quad & \tilde{y}_1 = 1 + 0.1(0 + 1) = 1.1, \\ & f(x_1, \tilde{y}_1) = 0.1 + 1.1 = 1.2, \\ \text{Corrector:}\quad & y_1 = 1 + \frac{0.1}{2}\left[1.0 + 1.2\right] = 1 + 0.05(2.2) = 1.11. \end{aligned}$$

Step 2 (from $(0.1,\ 1.11)$):

$$\begin{aligned} & f(x_1, y_1) = 0.1 + 1.11 = 1.21, \\ \text{Predictor:}\quad & \tilde{y}_2 = 1.11 + 0.1(1.21) = 1.231, \\ & f(x_2, \tilde{y}_2) = 0.2 + 1.231 = 1.431, \\ \text{Corrector:}\quad & y_2 = 1.11 + \frac{0.1}{2}\left[1.21 + 1.431\right] = 1.11 + 0.05(2.641) = 1.24205. \end{aligned}$$

Hence $y(0.2) \approx 1.2421$ (4 d.p.). Comparison: the exact value is 1.24281; the improved Euler error is about 8×10^{-4}, whereas plain Euler gives $y_2 = 1.22$ with error 2×10^{-2} — roughly 25 times worse for the same step. Averaging the gradients pays off handsomely.

7.8 Choosing a Method and Interpreting the Results

METHOD **Strategy for GCE Numerical Methods Questions**

1. Integrals: count the ordinates/strips; if the number of strips is even, Simpson's rule is expected. Quote the formula, show the ordinates, then substitute.
2. Series: decide between direct differentiation and using standard expansions; for products or composites, substitution and multiplication of known series is usually faster.
3. Differential equations: identify $f(x, y)$, x_0, y_0, h; write the recurrence before computing; present each step clearly.
4. Accuracy: smaller h gives smaller error; improved Euler beats Euler for the same h; Simpson beats the trapezium rule for the same n.
5. Always finish by stating the answer to the required accuracy, with units if the context demands them.

WORKED EXAMPLE 20 **Mixed GCE-style problem**

(a) Show that the Maclaurin expansion of $\ln(1 + x)$ up to the term in x^3 is $x - \frac{x^2}{2} + \frac{x^3}{3}$.

(b) Hence estimate $\int_0^{0.6} \frac{\ln(1 + x)}{x}\,dx$ by integrating the series term by term, giving 4 decimal places.

(c) Use Simpson's rule with 4 strips to estimate the same integral (take the value of the integrand at $x = 0$ to be its limit as $x \to 0$). Give your answer to 4 decimal places.

SOLUTION

Solution. (a) Let $f(x) = \ln(1 + x)$. Then

$$f'(x) = \frac{1}{1 + x}, \quad f''(x) = -\frac{1}{(1 + x)^2}, \quad f'''(x) = \frac{2}{(1 + x)^3},$$

so $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2$. Hence

$$\ln(1 + x) \approx 0 + x + \frac{-1}{2!}x^2 + \frac{2}{3!}x^3 = x - \frac{x^2}{2} + \frac{x^3}{3}.$$

(b) Dividing by x: $\frac{\ln(1 + x)}{x} \approx 1 - \frac{x}{2} + \frac{x^2}{3}$. Integrating term by term:

$$\int_0^{0.6} \frac{\ln(1 + x)}{x}\,dx \approx \left[x - \frac{x^2}{4} + \frac{x^3}{9}\right]_0^{0.6}$$

$$= 0.6 - \frac{0.36}{4} + \frac{0.216}{9} = 0.6 - 0.09 + 0.024 = 0.5340.$$

(c) Let $g(x) = \dfrac{\ln(1+x)}{x}$ with $g(0) = \lim_{x\to 0} \dfrac{\ln(1+x)}{x} = 1$ (from part (a), dividing by x and letting $x \to 0$). With $h = \dfrac{0.6}{4} = 0.15$:

$$g(0) = 1.000000, \qquad g(0.15) = \frac{\ln 1.15}{0.15} = 0.932508,$$
$$g(0.30) = \frac{\ln 1.3}{0.3} = 0.874547, \qquad g(0.45) = \frac{\ln 1.45}{0.45} = 0.825296,$$
$$g(0.60) = \frac{\ln 1.6}{0.6} = 0.783339.$$

Simpson's rule:

$$I \approx \frac{0.15}{3}\left[1.000000 + 0.783339 + 4(0.932508 + 0.825296) + 2(0.874547)\right]$$
$$= 0.05\left[1.783339 + 4(1.757804) + 1.749094\right]$$
$$= 0.05(10.563649) = 0.5282 \text{ (4 d.p.)}.$$

Comment: the series estimate 0.5340 uses only three terms of a series truncated at x^3; including the next term $-\frac{x^4}{4}$ in $\ln(1+x)$ contributes $-\frac{0.6^4}{16} = -0.0081$, giving 0.5259, much closer to Simpson's 0.5282. This illustrates both the power and the limitation of truncated Taylor approximations away from the origin.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 7 **MCQ: Simpson's rule**

Choose the correct answer, justifying your choice briefly.

1. Simpson's rule with n strips requires n to be:
   (a) any positive integer;
   (b) an even positive integer;
   (c) an odd positive integer;
   (d) a multiple of 3.
2. With strip width h and ordinates $y_0, y_1, \dots, y_n$, Simpson's rule states that $\int_a^b f(x)\,dx \approx$
   (a) $\frac{h}{2}\left[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\right]$;
   (b) $\frac{h}{3}\left[y_0 + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots) + y_n\right]$;
   (c) $\frac{h}{3}\left[y_0 + 2(y_1 + y_3 + \dots) + 4(y_2 + y_4 + \dots) + y_n\right]$;
   (d) $h\left(y_0 + y_1 + \cdots + y_n\right)$.

EXERCISE 8 **True or false**

State whether each statement is true or false, justifying your answer.

1. The Maclaurin series is the Taylor series of a function about $x = 0$.
2. Euler's method with step h uses the recurrence $y_{n+1} = y_n + h\,f(x_n, y_n)$.
3. Simpson's rule gives the exact value of $\int_a^b p(x)\,dx$ whenever p is a polynomial of degree at most 3.
4. The Maclaurin series of e^x converges only for $|x| < 1$.

EXERCISE 9 **Definitions**

1. Define the **Taylor series** of a function f about the point $x = a$, stating clearly the general term.
2. Explain, in one or two sentences, the geometric idea behind **Euler's method** for solving $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$.
3. State what is meant by the **improved Euler method** (Heun's method), giving its two-stage formula.

EXERCISE 10 **Direct calculations**

1. Write down the first four non-zero terms of the Maclaurin series of $\sin x$.
2. Given $f(x) = \sqrt{x}$, compute the ordinates needed to apply Simpson's rule with 4 strips to $\int_1^2 \sqrt{x}\,dx$ (take $h = 0.25$), but do not evaluate the estimate.
3. Given $\frac{dy}{dx} = x - y$ with $y(0) = 2$, perform one step of Euler's method with $h = 0.2$ to estimate $y(0.2)$.

8.2 I practise

EXERCISE 11 **Simpson's rule with 8 strips**

1. Use Simpson's rule with 8 strips to estimate $\int_0^2 \sqrt{1 + x^3}\,dx$, giving your answer to 3 decimal places. Present your ordinates in a table.
2. Explain why it is difficult to evaluate this integral exactly by elementary methods.

EXERCISE 12 **Cross-section of a canal**

The depths of a canal cross-section, measured at points every 3 m across the canal, are:

$$0,\quad 2.1,\quad 3.4,\quad 4.0,\quad 3.6,\quad 2.5,\quad 0 \quad \text{(metres)}.$$

1. Use Simpson's rule to estimate the area of the cross-section.

2. Hence estimate the volume of water in a 50 m length of the canal.
3. State one assumption you have made.

EXERCISE 13 **Maclaurin series of $\ln(1 + x)$**

1. Obtain the Maclaurin series of $\ln(1 + x)$ up to and including the term in x^4.
2. Use your series to estimate $\ln(1.2)$ to 4 decimal places.
3. Justify the accuracy of your estimate by considering the size of the first neglected term.

EXERCISE 14 **Limits via Taylor series**

Using appropriate Maclaurin series, evaluate:

1. $\lim_{x \to 0} \dfrac{e^x - 1 - x}{x^2}$;
2. $\lim_{x \to 0} \dfrac{x - \sin x}{x^3}$.

Explain why a direct substitution $x = 0$ fails in each case.

8.3 I prepare for the GCE

EXERCISE 15 **Integrating e^{-x^2}** **(12 marks)**

The integral $I = \displaystyle\int_0^1 e^{-x^2}\,dx$ cannot be expressed in terms of elementary functions.

(a) Write down the Maclaurin series of e^u and deduce the Maclaurin series of e^{-x^2} up to the term in x^8. (3 marks)

(b) By integrating your series term by term, estimate I to 4 decimal places, justifying the number of terms used. (4 marks)

(c) Use Simpson's rule with 4 strips to obtain a second estimate of I, giving your answer to 4 decimal places. (4 marks)

(d) Comment on the agreement between the two estimates. (1 mark)

EXERCISE 16 **Euler and improved Euler methods** **(14 marks)**

Consider the differential equation $\frac{dy}{dx} = x + y$ with $y(0) = 1$.

(a) Use Euler's method with step $h = 0.1$ to estimate $y(0.3)$, presenting all your working in a clearly labelled table. (4 marks)

(b) Repeat the calculation using the improved Euler method with the same step $h = 0.1$. (5 marks)

(c) Verify that $y = 2e^{x} - x - 1$ is the exact solution of the equation, and compute $y(0.3)$ to 5 decimal places. (3 marks)

(d) Compare the errors of the two numerical methods and comment on your findings. (2 marks)

EXERCISE 17 **GCE-style structured question** **(16 marks)**

(a) Derive the Maclaurin series of $\cos x$, giving the general term and the first three non-zero terms. (4 marks)

(b) Hence approximate $\cos(0.3)$ to 5 decimal places, and estimate the error involved. (4 marks)

(c) Use Simpson's rule with 6 strips to estimate $\int_{0}^{\pi} \cos x \, dx$, giving your answer to 5 decimal places. (5 marks)

(d) The exact value of the integral in part (c) is 0. Explain why the Simpson's rule estimate has a surprisingly small error, relating your answer to the symmetry of the cosine curve on $[0, \pi]$. (3 marks)

9 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Simpson's rule**

1. Correct answer: (b) an even positive integer.
Simpson's rule works on pairs of strips: a parabola is fitted through three consecutive ordinates $y_{2k}, y_{2k+1}, y_{2k+2}$, and this parabola spans exactly two strips of width h. The strips are therefore consumed two at a time, so the total number of strips must be even. (Equivalently, the number of ordinates must be odd.)
2. Correct answer: (b) $\frac{h}{3}\left[y_0 + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots) + y_n\right]$.
The pattern of weights is $1, 4, 2, 4, 2, \dots, 4, 1$: the two end ordinates have coefficient 1, the odd-indexed ordinates have coefficient 4, and the even-indexed interior ordinates have coefficient 2, the whole sum being multiplied by $\frac{h}{3}$. Answer (a) is the trapezium rule (weights $1, 2, 2, \dots, 2, 1$ with factor $\frac{h}{2}$), answer (c) swaps the coefficients 2 and 4, and answer (d) has no weighting at all.

SOLUTION **Exercise 2 — True or false**

1. True. By definition, the Maclaurin series of f is the Taylor series about $a = 0$:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

2. True. This is exactly Euler's recurrence: the slope $f(x_n, y_n)$ is treated as constant over the whole interval $[x_n, x_{n+1}]$, so the increment in y is $h\,f(x_n, y_n)$, giving $y_{n+1} = y_n + h\,f(x_n, y_n)$.
3. True. Simpson's rule is built by integrating interpolating parabolas, and its error term involves the fourth derivative $f^{(4)}$. For any polynomial of degree at most 3, $f^{(4)} \equiv 0$, so the error is exactly zero and the rule is exact. Notice that the statement even underestimates the rule: Simpson's rule is exact for cubics, not merely for quadratics.
4. False. The Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges for all real x (its radius of convergence is infinite). Indeed, by the ratio test, for any fixed x:

$$\left|\frac{x^{n+1}/(n+1)!}{x^n/n!}\right| = \frac{|x|}{n+1} \xrightarrow[n\to\infty]{} 0 < 1,$$

so the series converges whatever the value of x.

SOLUTION **Exercise 3 — Definitions**

1. The **Taylor series** of f about $x = a$ is

$$f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

with general term $\frac{f^{(n)}(a)}{n!}(x-a)^n$, where $f^{(n)}(a)$ denotes the n-th derivative of f evaluated at a (with the conventions $f^{(0)} = f$ and $0! = 1$).

2. In **Euler's method**, the solution curve through (x_0, y_0) is replaced, on each small interval of width h, by its tangent line at the current point: from (x_n, y_n) one moves along the direction of slope $f(x_n, y_n)$ to the next point (x_{n+1}, y_{n+1}), where $x_{n+1} = x_n + h$ and $y_{n+1} = y_n + h\,f(x_n, y_n)$. The true solution curve is thus approximated by a polygonal line (a sequence of straight segments).

3. The **improved Euler method** (Heun's method) is a two-stage predictor–corrector scheme:

$$\text{predictor:} \quad y^* = y_n + h\,f(x_n, y_n),$$
$$\text{corrector:} \quad y_{n+1} = y_n + \frac{h}{2}\Big[f(x_n, y_n) + f\big(x_{n+1}, y^*\big)\Big].$$

The predictor makes a first Euler estimate y^* of y_{n+1}; the corrector then advances using the average of the slopes at the beginning and at the (predicted) end of the interval, instead of the initial slope alone. This is why it is substantially more accurate than Euler's method for the same step size.

SOLUTION **Exercise 4 — Direct calculations**

1. Let $f(x) = \sin x$. The derivatives cycle with period 4:

$$f'(x) = \cos x, \quad f''(x) = -\sin x, \quad f'''(x) = -\cos x, \quad f^{(4)}(x) = \sin x,$$

so at $x = 0$ all even derivatives vanish ($\sin 0 = 0$) and the odd derivatives alternate in sign: $f'(0) = 1$, $f'''(0) = -1$, $f^{(5)}(0) = 1$, …. Substituting into the Maclaurin formula:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \ldots$$

2. With $h = 0.25$ on $[1, 2]$, the nodes are $x = 1,\ 1.25,\ 1.5,\ 1.75,\ 2$, and the ordinates of $y = \sqrt{x}$ are:

$$y_0 = 1, \quad y_1 = \sqrt{1.25} \approx 1.1180, \quad y_2 = \sqrt{1.5} \approx 1.2247, \quad y_3 = \sqrt{1.75} \approx 1.3229, \quad y_4 = \sqrt{2} \approx 1.4142.$$

3. Here $f(x, y) = x - y$, $x_0 = 0$, $y_0 = 2$, $h = 0.2$. Euler's recurrence gives

$$y_1 = y_0 + h\,f(x_0, y_0) = 2 + 0.2\,(0 - 2) = 2 - 0.4 = 1.6.$$

So $y(0.2) \approx 1.6$.

SOLUTION **Exercise 5 — Simpson's rule with 8 strips**

1. With 8 strips on $[0, 2]$, the strip width is $h = \dfrac{2 - 0}{8} = 0.25$. Computing the ordinates of $f(x) = \sqrt{1 + x^3}$ (kept to 5 d.p. to protect the final accuracy):

x	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2
$y = \sqrt{1 + x^3}$	1	1.00778	1.06066	1.19242	1.41421	1.71847	2.09165	2.52178	3

Grouping the ordinates according to the Simpson pattern 1, 4, 2, 4, 2, 4, 2, 4, 1:

$$\begin{aligned} \text{ends:} &\quad y_0 + y_8 = 1 + 3 = 4, \\ \text{odd indices:} &\quad y_1 + y_3 + y_5 + y_7 = 1.00778 + 1.19242 + 1.71847 + 2.52178 = 6.44045, \\ \text{even indices:} &\quad y_2 + y_4 + y_6 = 1.06066 + 1.41421 + 2.09165 = 4.56652. \end{aligned}$$

Therefore

$$\begin{aligned} \int_0^2 \sqrt{1 + x^3}\, dx &\approx \frac{0.25}{3}\Big[4 + 4(6.44045) + 2(4.56652)\Big] \\ &= \frac{0.25}{3}\Big[4 + 25.76180 + 9.13304\Big] = \frac{0.25}{3}(38.89484) \approx 3.241. \end{aligned}$$

2. The integrand $\sqrt{1 + x^3}$ is the square root of a cubic polynomial. No substitution or standard technique reduces it to elementary functions: its antiderivative can only be expressed in terms of elliptic integrals, which are non-elementary. Hence a numerical method such as Simpson's rule is the practical way to obtain a value.

SOLUTION **Exercise 6 — Cross-section of a canal**

1. There are 7 ordinates, i.e. 6 strips; since 6 is even, Simpson's rule applies, with spacing $h = 3$ m. The depths (in metres) are 0, 2.1, 3.4, 4.0, 3.6, 2.5, 0, so

$$\begin{aligned} A &\approx \frac{3}{3}\Big[(0 + 0) + 4(2.1 + 4.0 + 2.5) + 2(3.4 + 3.6)\Big] \\ &= 1 \times \Big[4(8.6) + 2(7.0)\Big] = 34.4 + 14 = 48.4 \text{ m}^2. \end{aligned}$$

2. Assuming the cross-section is uniform along the 50 m length of the canal:

$$V \approx 48.4 \times 50 = 2420 \text{ m}^3.$$

3. The main assumption is that the cross-sectional area (and shape) of the canal is constant along the whole 50 m length, so that the volume is simply (cross-sectional area) × (length). One also assumes that the depth varies smoothly between the measured points, so that the parabolic arcs used by Simpson's rule model the bed of the canal well.

SOLUTION **Exercise 7 — Maclaurin series of $\ln(1+x)$**

1. Let $f(x) = \ln(1+x)$. Differentiating successively:

$$f'(x) = \frac{1}{1+x}, \quad f''(x) = -\frac{1}{(1+x)^2}, \quad f'''(x) = \frac{2}{(1+x)^3}, \quad f^{(4)}(x) = -\frac{6}{(1+x)^4},$$

so $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2$, $f^{(4)}(0) = -6$. Substituting into the Maclaurin formula $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \dots$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \qquad \text{(valid for } -1 < x \le 1\text{)}.$$

2. Setting $x = 0.2$:

$$\ln(1.2) \approx 0.2 - \frac{0.04}{2} + \frac{0.008}{3} - \frac{0.0016}{4} = 0.2 - 0.02 + 0.0026667 - 0.0004 = 0.1822667.$$

Hence $\ln(1.2) \approx 0.1823$ (4 d.p.).

3. For $0 < x \le 1$ the series is alternating with terms decreasing in magnitude, so the error is bounded by the first neglected term:

$$\left|\frac{x^5}{5}\right| = \frac{0.2^5}{5} = \frac{0.00032}{5} = 0.000064.$$

This error is of order 10^{-4}: it can affect at most the fourth decimal place. The estimate 0.1823 is therefore reliable to 4 d.p., which is confirmed by the true value $\ln(1.2) = 0.18232\dots$

SOLUTION **Exercise 8 — Limits via Taylor series**

In both cases, direct substitution of $x = 0$ gives the indeterminate form $\frac{0}{0}$, because numerator and denominator both tend to zero. The Maclaurin series resolves the indeterminacy by revealing the rate at which each part tends to zero.

1. Since $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, the numerator is $e^x - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \dots$, so

$$\frac{e^x - 1 - x}{x^2} = \frac{\frac{x^2}{2} + \frac{x^3}{6} + \dots}{x^2} = \frac{1}{2} + \frac{x}{6} + \dots \xrightarrow[x\to 0]{} \frac{1}{2}.$$

2. Since $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, the numerator is $x - \sin x = \frac{x^3}{6} - \frac{x^5}{120} + \dots$, so

$$\frac{x - \sin x}{x^3} = \frac{\frac{x^3}{6} - \frac{x^5}{120} + \dots}{x^3} = \frac{1}{6} - \frac{x^2}{120} + \dots \xrightarrow[x\to 0]{} \frac{1}{6}.$$

SOLUTION **Exercise 9 — Integrating e^{-x^2} (12 marks)**

(a) (3 marks). The Maclaurin series of the exponential is

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \dots \qquad \text{(valid for all } u\text{)}.$$

Substituting $u = -x^2$:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \frac{x^8}{24} - \dots$$

Examiner's expectation: B1 for the series of e^u; M1 for the substitution $u = -x^2$; A1 for all signs and denominators correct up to x^8.

(b) (4 marks). Integrating term by term on $[0, 1]$:

$$I \approx \int_0^1 \left(1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \frac{x^8}{24}\right) dx = \left[x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \frac{x^9}{216}\right]_0^1$$

$$= 1 - 0.3333333 + 0.1 - 0.0238095 + 0.0046296 = 0.7474868 \approx 0.7475.$$

The series is alternating with decreasing terms on $[0, 1]$, so the error is bounded by the first neglected term. The next term of the series is $-\frac{x^{10}}{120}$, whose integral is $\frac{1}{11 \times 120} = \frac{1}{1320} \approx 0.00076$. Including this extra term gives the corrected estimate $I \approx 0.7474868 - 0.0007576 = 0.7467292 \approx 0.7467$, confirming that the value is stable to three decimal places (the true value is $0.74682\dots$).

Examiner's expectation: M1 for integrating term by term; A1 for the correct antiderivative values; A1 for 0.7475 (or 0.7467 with the extra term); B1 for an error justification via the first neglected term.

(c) (4 marks). With 4 strips on $[0, 1]$, $h = \frac{1 - 0}{4} = 0.25$:

x	0	0.25	0.5	0.75	1
e^{-x^2}	1	0.939413	0.778801	0.569783	0.367879

Applying Simpson's rule with the pattern 1, 4, 2, 4, 1:

$$I \approx \frac{0.25}{3}\left[(1 + 0.367879) + 4(0.939413 + 0.569783) + 2(0.778801)\right]$$

$$= \frac{0.25}{3}\left[1.367879 + 6.036784 + 1.557602\right] = \frac{0.25}{3}(8.962265) = 0.746855 \approx 0.7469.$$

Examiner's expectation: B1 for $h = 0.25$ and correct ordinates; M1 for the correct Simpson pattern (1, 4, 2, 4, 1); A2 for 0.7469 to 4 d.p.

(d) (1 mark). The two estimates, 0.7467 (series) and 0.7469 (Simpson), agree to within about 2×10^{-4}, and both match the true value 0.74682 to at least three decimal places. Two independent methods giving nearly the same value strongly confirms the accuracy of both.

Mark scheme summary: (a) 3, (b) 4, (c) 4, (d) 1 — total 12 marks.

SOLUTION **Exercise 10 — Euler and improved Euler methods (14 marks)**

Here $f(x,y) = x + y$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$.

(a) (4 marks). Euler's recurrence is $y_{n+1} = y_n + 0.1\,(x_n + y_n)$. Working step by step:

n	x_n	y_n	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.1f$
0	0	1	1	1.1
1	0.1	1.1	1.2	1.22
2	0.2	1.22	1.42	1.362

Hence $y(0.3) \approx 1.362$.

Examiner's expectation: M1 for the correct recurrence; A1 for each correct row (3 steps); A1 for the final value.

(b) (5 marks). Improved Euler: predict $y^* = y_n + hf_n$, then correct with $y_{n+1} = y_n + \frac{h}{2}\left[f_n + f(x_{n+1}, y^*)\right]$.

Step 1: $f_0 = 0 + 1 = 1$; $y^* = 1 + 0.1(1) = 1.1$; $f(0.1, 1.1) = 1.2$;

$$y_1 = 1 + 0.05(1 + 1.2) = 1.11.$$

Step 2: $f_1 = 0.1 + 1.11 = 1.21$; $y^* = 1.11 + 0.1(1.21) = 1.231$; $f(0.2, 1.231) = 1.431$;

$$y_2 = 1.11 + 0.05(1.21 + 1.431) = 1.11 + 0.13205 = 1.24205.$$

Step 3: $f_2 = 0.2 + 1.24205 = 1.44205$; $y^* = 1.24205 + 0.1(1.44205) = 1.386255$; $f(0.3, 1.386255) = 1.686255$;

$$y_3 = 1.24205 + 0.05(1.44205 + 1.686255) = 1.24205 + 0.1564153 = 1.3984653.$$

Hence $y(0.3) \approx 1.39847$.

Examiner's expectation: M1 for the predictor–corrector structure; A1 per correct step (3); A1 for the final value 1.39847.

(c) (3 marks). Let $y = 2e^x - x - 1$. Check the two conditions:

Initial condition: $y(0) = 2e^0 - 0 - 1 = 2 - 1 = 1$, as required.

Differential equation:

$$\frac{dy}{dx} = 2e^x - 1 = (2e^x - x - 1) + x = y + x,$$

so y satisfies $\frac{dy}{dx} = x + y$. Since y satisfies both the equation and the initial condition, it is the exact solution. Therefore

$$y(0.3) = 2e^{0.3} - 0.3 - 1 = 2(1.3498588) - 1.3 = 1.3997176 \approx 1.39972 \quad (5\text{ d.p.}).$$

Examiner's expectation: B1 for checking the initial condition; B1 for checking the differential equation; B1 for 1.39972.

(d) (2 marks). Errors at $x = 0.3$:

$$\text{Euler:} \quad |1.39972 - 1.362| = 0.03772,$$
$$\text{Improved Euler:} \quad |1.39972 - 1.39847| = 0.00125.$$

The improved Euler error is about 30 times smaller. This is expected: Euler's method is first-order (global error proportional to h), whereas the improved Euler method is second-order (global error proportional to h^2), because averaging the slopes at the two ends of the interval follows the curvature of the solution far better than using only the initial slope.
Mark scheme summary: (a) 4, (b) 5, (c) 3, (d) 2 — total 14 marks.

SOLUTION **Exercise 11 — GCE-style structured question (16 marks)**

(a) (4 marks). Let $f(x) = \cos x$. The derivatives cycle with period 4:

$$f'(x) = -\sin x, \quad f''(x) = -\cos x, \quad f'''(x) = \sin x, \quad f^{(4)}(x) = \cos x.$$

At $x = 0$: $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$, $f^{(4)}(0) = 1$, and in general $f^{(2n)}(0) = (-1)^n$ while $f^{(2n+1)}(0) = 0$. Substituting into the Maclaurin formula:

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

The general term is $\dfrac{(-1)^n x^{2n}}{(2n)!}$; the first three non-zero terms are $1 - \dfrac{x^2}{2} + \dfrac{x^4}{24}$.
Examiner's expectation: M1 for differentiating and evaluating at 0; A1 for the pattern of values; B1 for the general term; A1 for the three terms.
(b) (4 marks). With $x = 0.3$:

$$\cos(0.3) \approx 1 - \frac{0.09}{2} + \frac{0.0081}{24} = 1 - 0.045 + 0.0003375 = 0.9553375 \approx 0.95534 \quad (5\text{ d.p.}).$$

The series is alternating with terms decreasing in magnitude, so the error is bounded by the first neglected term:

$$\left|\frac{x^6}{6!}\right| = \frac{0.3^6}{720} = \frac{0.000729}{720} \approx 1.0 \times 10^{-6},$$

which cannot affect the fifth decimal place.
Examiner's expectation: M1 for the substitution; A1 for 0.95534; M1, A1 for the error bound $\approx 10^{-6}$.
(c) (5 marks). With 6 strips on $[0, \pi]$, $h = \dfrac{\pi - 0}{6} = \dfrac{\pi}{6}$. The ordinates of $\cos x$ are exact standard values:

x	0	$\pi/6$	$\pi/3$	$\pi/2$	$2\pi/3$	$5\pi/6$	π
$\cos x$	1	0.86603	0.5	0	-0.5	-0.86603	-1

Applying Simpson's rule with the pattern 1, 4, 2, 4, 2, 4, 1:

$$\int_0^{\pi} \cos x\,dx \approx \frac{\pi/6}{3}\left[(1 + (-1)) + 4(0.86603 + 0 - 0.86603) + 2(0.5 - 0.5)\right]$$
$$= \frac{\pi}{18}\left[0 + 4(0) + 2(0)\right] = 0.00000.$$

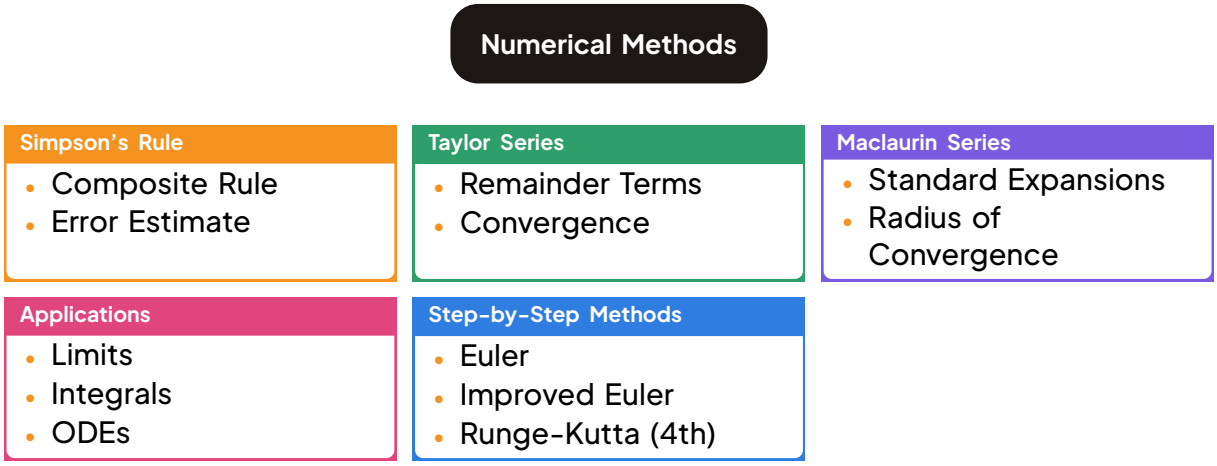
The Simpson estimate is exactly 0.00000 (which agrees with the exact value $[\sin x]_0^{\pi} = 0$). Examiner's expectation: B1 for $h = \pi/6$; B1 for the ordinates; M1 for the Simpson pattern; A2 for the value 0.

(d) (3 marks). The cosine curve on $[0, \pi]$ has point symmetry about $\left(\frac{\pi}{2}, 0\right)$: $\cos(\pi - x) = -\cos x$. The ordinates therefore occur in cancelling pairs: $y_0 = -y_6$, $y_1 = -y_5$, $y_2 = -y_4$, and $y_3 = 0$. Moreover, the Simpson weights respect this symmetry, since symmetrically placed ordinates carry equal coefficients ($1 \leftrightarrow 1, 4 \leftrightarrow 4, 2 \leftrightarrow 2$). Consequently the positive and negative contributions cancel exactly in the weighted sum, giving an error of precisely zero — not merely a small error. In short, the symmetry of the integrand is inherited exactly by the quadrature formula.

Examiner's expectation: B1 for stating the symmetry $\cos(\pi - x) = -\cos x$; B1 for noting that the Simpson weights are symmetric; B1 for concluding exact cancellation.

Mark scheme summary: (a) 4, (b) 4, (c) 5, (d) 3 — total 16 marks.

10 SUMMARY SHEET



Mind map of Chapter FMA10: Numerical Methods

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **Simpson's Rule**: Numerical integration using quadratic interpolation over pairs of subintervals. Requires an even number of strips n.
- **Taylor Series**: Expansion of $f(x)$ about $x = a$: $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$.
- **Maclaurin Series**: Taylor series about $a = 0$: $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k$.
- **Remainder (Lagrange form)**: $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$ for some ξ between a and x.
- **Step-by-step methods**: Iterative algorithms to approximate solutions of $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$ at discrete points $x_{r+1} = x_r + h$.

Laws, Formulas, Theorems (Examinable Proofs Marked ★)

- Simpson's Rule (Composite):

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[y_0 + y_n + 4\sum_{\text{odd } i} y_i + 2\sum_{\text{even } i} y_i\right], \quad h = \frac{b-a}{n},\ n \text{ even}.$$

- Error in Simpson's Rule (★ proof not required, formula must be known):

$$E_S = -\frac{(b-a)h^4}{180}f^{(4)}(\xi), \quad \xi \in (a, b).$$

- Taylor's Theorem (★ proof examinable): If $f \in C^{n+1}[a, x]$, then $f(x) = P_n(x) + R_n(x)$ with P_n the Taylor polynomial and R_n the remainder.
- Maclaurin Expansions (must be memorised):

$$e^x = \sum \frac{x^k}{k!},\ \sin x = \sum \frac{(-1)^k x^{2k+1}}{(2k+1)!},\ \cos x = \sum \frac{(-1)^k x^{2k}}{(2k)!},\ \ln(1+x) = \sum \frac{(-1)^{k-1} x^k}{k},\ (1+x)^\alpha = \sum \binom{\alpha}{k} x^k.$$

- Euler's Method: $y_{r+1} = y_r + hf(x_r, y_r)$; local truncation error $O(h^2)$, global $O(h)$.
- Improved Euler (Heun):

$$k_1 = f(x_r, y_r), \quad k_2 = f(x_r + h,\ y_r + hk_1), \quad y_{r+1} = y_r + \frac{h}{2}(k_1 + k_2).$$

- Runge-Kutta 4th Order:

$$\begin{aligned} k_1 &= f(x_r, y_r), \\ k_2 &= f(x_r + \tfrac{h}{2}, y_r + \tfrac{h}{2}k_1), \\ k_3 &= f(x_r + \tfrac{h}{2}, y_r + \tfrac{h}{2}k_2), \\ k_4 &= f(x_r + h, y_r + hk_3), \\ y_{r+1} &= y_r + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

  Global error $O(h^4)$.

Methods (Step-by-Step)

1. Simpson's Rule: (1) Choose even n, compute h. (2) Tabulate x_i, $y_i = f(x_i)$. (3) Apply weighting pattern $1, 4, 2, 4, \ldots, 2, 4, 1$. (4) Multiply sum by $h/3$. (5) Estimate error using $f^{(4)}$ bound if required.
2. Taylor/Maclaurin Expansion: (1) Compute derivatives at centre. (2) Write polynomial up to required order. (3) State remainder term. (4) For standard functions, use known series and substitute/combine.
3. Approximating Integrals/Limits: Replace integrand/function by Taylor polynomial, integrate/expand term by term, use remainder to bound error.
4. Solving ODEs: (1) Identify $f(x, y)$, h, initial (x_0, y_0). (2) Apply chosen scheme iteratively. (3) Tabulate results. (4) Compare methods: smaller h or higher order improves accuracy.

Common Mistakes

- Using Simpson's rule with odd n (must be even).
- Forgetting the factor $h/3$ or misapplying the $4/2$ weighting pattern.

- Confusing Taylor remainder forms; Lagrange form is standard for error bounds.
- Omitting the factorial denominators in series coefficients.
- Assuming a Taylor series converges to the function everywhere; always check interval of convergence (ratio test).
- In step-by-step methods: mixing up x and y updates, or using k_1 instead of k_2 in Improved Euler's predictor step.
- Forgetting that Euler's method overestimates/underestimates depending on concavity (sign of f_y or y'').

GCE Tips

- Simpson's Rule: Questions often ask to derive the formula for a single strip (★) or apply composite rule with given table. Always show the weighting pattern clearly.
- Taylor/Maclaurin: "Find the expansion up to x^4" means compute derivatives up to 4th order. "State the range of validity" requires ratio test or known radius.
- Error Estimation: Use the Lagrange remainder to find n such that $|R_n| < \epsilon$, or to bound the error of a numerical integral.
- ODEs: You may be asked to perform 2–3 iterations of Euler/Improved Euler/RK4 by hand. Keep a clear table: r, x_r, y_r, k values, y_{r+1}. Show all k calculations for RK4.
- Comparison: "Explain why Runge-Kutta is more accurate than Euler" — mention order of local truncation error (h^5 vs h^2) and use of weighted slopes.
- Calculator Use: In the exam, you may use a calculator for arithmetic, but you must show the formula substitution and method steps to gain method marks.

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